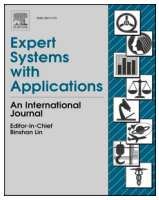
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A multitasking multi-objective differential evolution gene selection algorithm enhanced with new elite and guidance strategies for tumor identification

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ABSTRACT

A key preprocessing step in tumor recognition based on microarray expression profile data and machine learning is to identify tumor marker genes. Gene selection aims to select the most relevant gene subset from the original ultra-high dimensional microarray expression profile data to improve tumor identification performance. Inspired by evolutionary multitasking (EMT) and multi-objective optimization, this paper puts forward a novel multitasking multi-objective differential evolution gene selection algorithm (MMODE) which uses new elite and guidance strategies to select the best gene subsets. MMODE initializes two different populations according to different filtering criteria to increase the diversity of the search. These two populations guide their respective populations to search in the optimal direction through knowledge transfer in the evolutionary process. In addition, MMODE employs new elite and guidance strategies that enables individuals to narrow the search range and jump out of local optima. The proposed algorithm is validated on 13 publicly available microarray expression datasets in comparison with state-of-the-art gene selection algorithms. The experimental results show that MMODE can find smaller gene subsets and achieve higher classification accuracy compared with other algorithms.

1. Introduction

The emergence and rapid development of large-scale gene expression profiling provide a new technology platform for tumor research. The features in microarray gene expression profile data are called genes, and the process of selecting a subset of features from these data that are most relevant for disease diagnosis is called gene selection. Tumor gene selection based on microarray expression profile data is a hot and difficult topic in the field of machine learning and data mining. Microarray gene expression profile data are characterized by high dimensionality (tens of thousands) and a small number of samples (dozens of), but few genes are related to disease diagnosis. How to identify the most relevant gene subsets from ultra-high dimensional microarray expression profile data to improve the accuracy of tumor recognition, which brings great challenges to the task of gene selection (Deng, Li, Deng, & Wang, 2022; Ke et al., 2022; Liang et al., 2022; Li et al., 2023).

Classification of high-dimensional data usually requires dimensionality reduction to improve the generalization of classification models and reduce the consumption of computational resources. As a data preprocessing technology, feature selection (FS) (Ray, Reddy, & Banerjee, 2021) reduces the number of features by removing irrelevant and redundant features, so as to achieve the purpose of improving the classification performance of learning model Chen, Yuan, Lv, & Xiang, 2021; Wang et al., 2017; Wang, Zhu, Li, Chen, & Zhang, 2021). Compared with feature extraction, FS can retain the physical meanings of the original feature space and improve the readability and interpretability of the learning model (Nie, Wang, Tian, Wang, & Li, 2022). Due to its effectiveness, FS has been widely used in data mining, such as cancer detection (Zibakhsh & Abadeh, 2013) and shop floor scheduling problems (Zhang, Mei, Nguyen, & Zhang, 2021).

To date, many FS methods have been developed, which can be roughly divided into the filter, wrapper, and hybrid methods (Hancer, Xue,& Zhang, 2022; Li, Shang, Feng, & Fan, 2014; Tiwari & Chaturvedi, 2022). The filter-based approaches searched for a subset of features based on characteristics of the features (e.g. similarity, correlation, etc.) (Ba, Wang, Yang, Yu, & Yu, 2023). The wrapper-based methods utilize a

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classification model to evaluate feature subsets in a cross-validation manner to determine the performance of candidate feature subsets (Abdollahzadeh & Gharehchopogh, 2022). Therefore, the wrapper-based methods are computationally expensive. In contrast, filter-based methods are less costly but often less accurate than wrapper-based methods. On combine the advantages of filer-based and wrapper-based methods, filter-wrapper-based hybrid methods were proposed (Ghosh, Guha, Sarkar, & Abraham, 2020). The filter-wrapper-based methods achieve high classification accuracy with low time consumption, which has a great advantage over filter-based and wrapper-based methods. Therefore, filter-wrapper-based methods are widely used in FS for high-dimensional data (Huang, Yang, Zhou, & Huang, 2019). Typically, a filter-wrapper-based approach includes two stages. The first stage uses a filter method to evaluate the importance of features and remove the most irrelevant and redundant features. Therefore, the results of the first stage are influenced by a filter being adopted. The second stage uses a wrapper-based approach to select the optimal feature subset from the remaining features, which belongs to a combinatorial optimization problem (Got, Moussaoui, & Zouache, 2021), and many swarm intelligence methods have been developed to solve this problem in recent years.

Differential evolution (DE), which was developed by Storn and Price (1997), is one of the most effective evolutionary algorithms (Wang, Cai, & Zhang, 2011). So far, many DE-based FS methods have been proposed. For example, a hybrid FS method based on sine cosine algorithm (SCA) and DE was proposed by Abd Elaziz et al. (2017). This method uses differential evolution operator to improve the local search strategy and overcomes the shortcoming that SCA is easy to fall into the local optimum. The history-guided differential evolution (HGDE) algorithm was recently proposed by Guan et al. (2021) for feature selection of high-dimensional data. It can remove duplicate individuals to maintain population diversity and guide the search direction of individuals according to search history, thus reducing the probability of falling into local optimum. In fact, many scholars have focused on solving the problem that DE-based FS algorithms are prone to fall into local optima (i.e., individuals are stagnant) (Liu et al., 2019; Wang, Xue, Liang, & Zhang, 2021; Wang et al., 2015). However, for high-dimensional microarray gene expression profile data, it is a very difficult task to find the optimal solution due to its huge gene subset search space. In addition, FS is considered a multi-objective problem since its two objectives are to reduce the number of features and to improve the classification performance. How to find the optimal subset of genes from microarray expression data still needs further study.

Multi-objective evolutionary algorithms (MOEAs) can obtain groups of non-dominated solutions and are frequently used to solve multi-objective optimization problems (MOPs). At present, MOEAs can be divided into three categories, namely, dominance-based algorithms (Deb, Pratap, Agarwal, & Meyarivan, 2002; Palakonda & Mallipeddi, 2017; Tian et al., 2019), decomposition-based algorithms (Han, Chen, Ling, & Han, 2021; Liao, Ishibuchi, & Meng Pang, 2020; Nguyen et al., 2020), and indicator-based methods (Bader & Zitzler, 2011; Hong et al., 2019; Zitzler & Künzli, 2004). However, these algorithms are also prone to fall into local optimum and premature convergence. In previous MOEAs, only one population is maintained during the population evolution. The disadvantage of this is that the population potentially lacks diversity since the population tends to converge to a locally optimal position in a multi-objective optimization problem (Zhao, Liu, Zhang, & Liu, 2019). Compared with single-population evolution, multiple-populations increase the diversity of populations, make individuals more evenly distributed, and effectively overcome the local optimum problem. Note that MOEAs cannot solve multiple MOPs simultaneously. In other words, MOEAs can only solve one multi-objective problem at a time. However, multiple MOPs are interrelated with each other, meaning that the beneficial knowledge gained in optimizing one MOPs can be used to solve other related problems.

To effectively cope with the above problem, Gupta et al. (2016) proposed a new paradigm called evolutionary multitasking (EMT) to handle multiple optimization tasks simultaneously. Due to the potential connection between task populations, the EMT algorithm often makes it easier to achieve an optimum solution than the corresponding single-task population algorithm. The EMT paradigm has attracted much attention from MOPs researchers. For example, a multi-objective multifactorial evolutionary algorithm MO-MFEA (Gupta, Ong, Feng, & Tan, 2017) was proposed based on nondominated sorting genetic algorithm II (NSGA-II) (Deb, Pratap, Agarwal, & Meyarivan, 2002). On this basis, Gupta proposed a new EMT algorithm, MO-MFEA-II (Bali, Gupta, Ong, & Tan, 2021), which improved the lack of information exchanged between tasks. In addition, Liang et al. (2022) proposed a new subspace alignment and adaptive DE algorithm for EMT, namely MOMFEA-SADE. This algorithm uses the mapping of subspaces to transform the search space of populations and reduce the probability of negative knowledge transfer between tasks. Recently, the EMT algorithm has also been applied to feature selection. For example, Chen et al. (2021) proposed a PSO-based multi-task algorithm that can find a small feature subset in a reasonable time and achieve high classification accuracy. However, to the best of our knowledge, there are very few FS applications of the EMT algorithm for high-dimensional data, which means that there is still research work to be done in this area.

Inspired by the EMT paradigm for solving complex problems, this paper designs a novel Multitasking Multi-objective Differential Evolution gene selection algorithm (MMODE) which uses new elite and guidance strategies to select the best gene subsets. MMODE searches the optimal gene subset by knowledge transfer between two related FS task populations. Different from the traditional evolutionary multi-task involving different problems, two FS tasks in this study are composed of different feature weights in various filter criteria, and the different weights of features determine the different search directions of each task. Two FS tasks have the same search space but not the same search direction, which ensures the internal relationship of the tasks and facilitates the knowledge transfer between tasks. In this multitasking system, knowledge is transferred between two tasks so that the population of one task can well guide the search for another task. In addition, to address the stagnation of individuals, a new guidance strategy is developed to cope with this situation. Furthermore, a new elitist strategy is adopted to reduce the search range, which effectively improves the performance of the algorithm for feature selection of high-dimensional data. For gene selection of microarray gene expression profile data, MMODE is promising to achieve high classification accuracy with a few genes. Specifically, the main contributions of this paper are outstanding as follows.

1) The MMODE proposed in this paper is a new feature selection algorithm based on multi-task DE algorithm.
2) MMODE generates differentiated task populations, transfers knowledge in the process of population evolution, adaptively changes mutation operators, and effectively enhances the diversity of populations.
3) Different from the traditional multi-objective paradigm, MMODE uses a new elite strategy and a new guidance strategy to reduce the search space and avoid falling into local optima. In the elite strategy, a non-dominated ranking is used to generate elite populations for multi-task evolution. The proposed guidance strategy can help stagnant individuals jump out of local optima.
4) The effectiveness of MMODE was validated by comparing it with a variety of state-of-the-art algorithms on 13 gene microarray datasets. Experimental results show that MMODE outperforms the comparison algorithms in terms of classification accuracy and number of selected features.

The rest of the paper is organized as follows. Section 2 describes the related work. Section 3 presents our proposed algorithm in detail. Section 4 gives the experimental results and analysis. Finally, Section 5

concludes the whole paper and proposes the next steps.

2. Related work

In this section, some related knowledges are briefly reviewed. Section 2.1 introduces the filter method used by MMODE. Section 2.2 introduces the differential evolution algorithm. In Section 2.3, the formal definition of multi-objective optimization is given. Section 2.4 presents evolutionary multitasking for multi-objective optimization.

2.1. Filter criteria

FS of high-dimensional data is extremely challenging due to a large number of redundant and irrelevant features, so promising subsets of features need to be identified quickly (i.e., a subset needs to include important features) to improve efficiency. Filter methods are popular due to their ability to extract a promising subset of features simply and quickly. Among them, ReliefF (Spolaor, Cherman, Monard, & Lee, 2013) and mutual information (Hancer, Xue, & Zhang, 2018) are widely used in the field of data mining because they can effectively determine the relationship between features and classes. In this study, ReliefF and mutual information are used to assign different weights to candidate features of the two task populations.

2.1.1. ReliefF

The Relief (Kononenko, 1994) algorithm is a nearest-neighbor-based feature evaluation algorithm, which is widely used to solve binary classification problems. ReliefF is an extension of Relief to handle multi-classification problems. According to ReliefF, the weights of features are evaluated based on their ability to distinguish the sampled instances from their neighbors in the same class and neighboring instances in different classes. The pseudo-code for calculating feature weights using ReliefF is given in Algorithm 1. In Algorithm 1, R_i is a selected sample, the h nearest neighbors of R_i in the same and different categories are denoted as $H_j(1,2,\cdots,h)$ and $M_j(1,2,\cdots,h)$, respectively. *Eq.*(1) is used to calculate the weight of each feature to discriminate between the sample and its neighbors. In short, the higher the weight, the more important the feature.

$$W_r[F_a] = W_r[F_a] - \sum_{j=1}^{h} \frac{f(F_a, R_i, H_j)}{u^* h} + \sum_{c \notin s(R)} \frac{\left[\frac{p(c)}{1-p(s(R))} * \sum_{j=1}^{h} f(F_a, R_i, M_j(c))\right]}{u^* h} \tag{1}$$

where W_r denotes a vector that is used to store the ReliefF values of all features, the subscript r represents the relief, $W_r[F_a]$ is the (ReliefF) weight of the feature F_a, h is the number of nearest neighbor samples, $s(R)$ is the class label of the sample R, $p(c)$ is the prior probability of class c, and $f(F_a, R_i, R_j)$ implies the difference between samples R_i and R_j on feature F_a.

With ReliefF algorithm, the weight of each feature can be calculated, and the ranking of features can be obtained accordingly, and then a subset of features containing a small number of top-ranking features can be selected. Therefore, using the ReliefF algorithm requires a threshold to determine the number (or proportion) of selected features. Too small a threshold will cause information loss, and too large a threshold will contain too much redundancy and noise features. For high-dimensional data sets, the top 3–5 % features are usually selected as the final feature subset. Microarray gene expression profile data contain tens of thousands of features. In this study, the threshold value is 3 %, which can filter out most irrelevant features and retain enough features as the optimal feature subset search space.

Algorithm 1: Pseudocode of ReliefF

Input: D, training dataset;u, the number of iterations; h, the number of nearest neighbor samples; p, number of features;
Output: W_r, the weighs of features;
1: Set $W_r = \emptyset, a = 1,2,\cdots,p$;
2: for $i = 1$ to u **do**
3: Randomly select a sample from D, denotes as R_i;
4: Find h nearest neighbors from the same category as sample R_i, records as H_i;
5: Find h nearest neighbors different from category R_i, denotes as $M_j(c)$; //
6: **for** $a = 1$ to p **do**
7: Update the weight $W_r[a]$ based on *Eq.*(1);
8: **end**
9: end
10: return Feature weight vector W_r.

2.1.2. Mutual information

Mutual information can be used to evaluate the degree of dependence between variables and to measure the similar uncertainty of one variable to another given variable.

Let X be a discrete random variable that takes on values from $\{x_1, x_2, \cdots, x_k\}$, $p(x_k)$ is the probability function. The information entropy $H(X)$ is defined as:

$$H(X) = -\sum_{k} p(x_k) log_2 p(x_k) \tag{2}$$

The joint entropy $H(X,Y)$ of two random variables $= \{x_1, x_2, \cdots, x_k\}$ and $Y = \{y_1, y_2, \cdots, y_z\}$ are defined as follows.

$$H(X,Y) = -\sum_{k,z} p(x_k, y_z) log_2 p(x_k, y_z) \tag{3}$$

On such basis, the mutual information $I(X;Y)$ is defined as *Eq.*(4).

$$I(X;Y) = H(X) + H(Y) - H(X,Y) \tag{4}$$

Mutual information $I(X;Y)$ is a measure of the degree of interdependence between random variables X and Y. To evaluate the relationship between features and class variable, X represents feature variables and Y represents class variables in feature selection.

2.2. Differential evolution

Differential evolution (DE) process consists of three operators: mutation operator, crossover operator and selection operator. Mutation operator creates mutation vectors V based on X vector of current parent population. *Eq.*(5) and *Eq.*(6) are two commonly used mutation strategies.

$$V_{i,t} = X_{r1,t} + F \bullet (X_{r2,t} - X_{r3,t}) \tag{5}$$

$$V_{i,t} = X_{i,t} + F \bullet \left(X^{p}_{best,t} - X_{i,t}\right) + F \bullet (X_{r1,t} - X_{r2,t}) \tag{6}$$

where F and $X^{p}_{best,t}$ respectively represent the scale factor between $(0,1)$ and one of the top p% individuals in the current (t th) generation population chosen (Zhang & Sanderson, 2009). $X_{i,t}$ denotes the i th individual in the parent population. $X_{rj,t}$ denotes a randomly selected individual, $rj \in \{1,2,\cdots,N\}$, $rj \neq i$, and $j = \{1,2,3\}$.

In general, the naming paradigm for various mutation strategies are

$DE/x/y$, where DE stands for differential evolution, x stands for the strategy to be executed by the denoted target vector X, and y is the number of differential vectors perturbing X. Therefore, $Eq.(5)$ and $Eq.(6)$ are called $DE/rand/1$ and $DE/current-to-pbest/1$, respectively.

Different mutation operators have different perturbations to the vector $X_{i,t}$. These perturbations affect the movement of the vector $X_{i,t}$. In $Eq.(5)$, the vector $X_{i,t}$ move in a random direction for exploring a larger space. The mutation operator ($Eq.(5)$) focuses on improving the global exploration capability of the algorithm. In $Eq.(6)$, the vector $X_{i,t}$ moved towards the best vector $X^p_{best,t}$ and there is a perturbation in the random direction, allowing the vector $X_{i,t}$ to explore the region around the vector $X^p_{best,t}$($X^p_{best,t}$ comes from archive A (Zhang, & Sanderson, 2009), which stores the top p% of outstanding individuals in parent population). The mutation operator ($Eq.(6)$) focuses on improving the local exploitation capability of the algorithm.

The mutation vector $V_{i,t}$ after the mutation operation is crossed with the target vector $X_{i,t}$ to obtain the final trial vector $U_{i,t}$, as shown in the following $Eq.(7)$.

$$U_{i,t,j} = \begin{cases} V_{i,t,j} if (rand(0,1) \le CRorj = j_{rand}) \\ X_{i,t,j} otherwise \end{cases} \tag{7}$$

where CR is the crossover rate between $[0,1]$, $j_{rand} \in \{1,2,\cdots,D\}$ is a random integer that ensures that at least one value in the generated experimental vector is from the V vector.

The selection operation selects (as shown in $Eq.(8)$) the better one from the target vector X and the trial vector U to survive into the next generation based on its fitness value $f(\bullet)$. The selection operator is shown below:

$$X_{i,t+1} = \begin{cases} U_{i,t} iff(U_{i,t}) < f(X_{i,t}) \\ X_{i,t} otherwise \end{cases} \tag{8}$$

where $f(U_{i,t})$ represents the fitness function value of $U_{i,t}$.

The above operations are repeated until the iteration stop condition is satisfied.

2.3. Multi-objective optimization

The goal of solving a multi-objective optimization problem (MOPs) is to find a set of tradeoff solutions that satisfy all conflicting objectives. The formal description of the multi-objective optimization problem is as follows.

For a search space of $X \subset \mathbb{R}^D$, a multi-objective optimization problem can be denoted in the following form

$$minimize F(x) = (f_1(x), f_2(x), \cdots, f_M(x)) \tag{9}$$

where $f_1(x), f_2(x), \cdots, f_M(x)$ are M different minimization objectives, $F(x)$ is a vector of objectives, and $x \in X$ is the decision vector. By the concept of Pareto domination, if $\forall i \in \{1,2,\cdots,M\} : f_i(x_1) \le f_i(x_2)$ and $\exists j \in \{1,2,\cdots,M\} : f_j(x_1) < f_j(x_2)$, then it is means that the solution x_1 dominates the solution x_2 (i.e. $x_1 \prec x_2$). To summarize, x^* is Pareto optimal if there are no solutions dominate x^*. The Pareto-front is the Pareto optimal set of mutually non-dominated x^*. Feature selection is considered a multi-objective problem since its two main objectives are usually conflicting, in which the number of features selected and the classification error rate needs to be minimized. The classification error rate is more important than the number of features selected, since the main task for feature selection is to reduce the classification error rate.

As with most multi-objective FS algorithms, the objective function for this experiment uses the classification error rate and the number of features selected, two objectives that are minimal optimization problems for this experiment. Specifically, these two objectives are defined as:

$$f_1(X_i) = 1 - Acc(X_i) \tag{10}$$

$$f_2(X_i) = \frac{\sum_{j=1}^{n} X_{ij}}{n} \tag{11}$$

where X_i represents the i th individual in the population; X_{ij} is either equal to 1 or 0, $X_{ij} == 1$ implies that the j th feature is selected by individual X_i, and $X_{ij} == 0$ implies that the j th feature is not selected by individual X_i; n represents the number of features; $Acc(X_i)$ represents the classification accuracy of that individual; $f_1(X_i)$ is the classification error rate of the individual X_i, and $f_2(X_i)$ represents the proportion of the number of features selected by that individual.

2.4. Evolutionary multitasking for multi-objective optimization

Most of the existing evolutionary algorithms are used to deal with a single task (including single-objective optimization problems and multi-objective optimization problems), with less research into solving multiple tasks simultaneously. Gupta et al. (2016) applied to multitask learning to evolutionary computing and proposed a paradigm evolutionary multitasking (EMT) for evolutionary computing. In evolutionary multitask learning, multiple tasks can transfer knowledge to each other when dealing with similar problems. The knowledge of one task is transferred to the target task, and the performance of the target task is improved. This kind of interactive learning between tasks is called transfer learning. Multi-task learning improves the performance of all tasks based on the inherent relationships between tasks, each task benefits from the knowledge gained by the other tasks.

EMT algorithms tend to obtain better solutions for the optimization tasks than the counterpart single-task-based algorithms and achieved success in various optimization problems including MOPs. Multi-objective multifactorial evolutionary algorithm (MO-MFEA) (Gupta, Ong, Feng, & Tan, 2017) as an approach to solving MOPs, which solves MOPs simultaneously using the knowledge gained from each task. In the MO-MFEA framework, each task T_k will be considered as a factor affecting the individual in the search process, while the individuals in the task population P^k are encoded by a unique method that facilitates interactive problem solving. Thus, an individual X_i of the population P^k can be denoted as $P^k(X_i)$ which can be represented as a solution in a special decoding way.

To allow all tasks to transfer knowledge with each other, a unified encoding is used to control the search scope of all tasks in the same search space. The implicit connection can be generated by all tasks being in the uniform search space. Transferring knowledge between tasks is determined by the parameter of random mating probability (RMP), which can be set to a value close to 1 when the correlation between two tasks is high, and a low value (usually close to 0) when the correlation between two tasks is low. This means that MO-MFEA is strongly dependent on the experimenter's prior knowledge about the tasks between them. To overcome this deficiency, Gupta et al. proposed MO-MFEA-II which generates the RMP matrix ($Eq.(12)$) based on the adaptive learning of the probability distribution of the respective task population.

$$\mathrm{RMP_{K,K}} = \begin{bmatrix} \mathrm{rmp_{1,1}} & \mathrm{rmp_{1,2}} & \cdots & \mathrm{rmp_{1,K}} \\ \mathrm{rmp_{2,1}} & \mathrm{rmp_{2,2}} & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \mathrm{rmp_{K,1}} & \cdots & \cdots & \mathrm{rmp_{K,K}} \end{bmatrix}, \tag{12}$$

where $\mathrm{rmp_{i,j}}\ (\mathrm{i,j} = 1,2,\cdots,\mathrm{K})$ represents the random mating probability between task I and task II, which can effectively control the knowledge transfer between the tasks. The basic framework of a multitask has K tasks. The K th task is denoted T_k, and the corresponding fitness function is f_k. Therefore, MO-MFEA-II can adaptively determine the RMP parameters between the two tasks, thus effectively controlling the knowledge transfer between tasks.

In summary, although some scholars are committed to the research of multi-tasking framework to solve the multi-objective optimization

	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8
Wr	2	1	3.2	1.2	1.3	0.4	0.3	1.7
R_normalize	0.49	0.32	0.7	0.3551	0.372	0.217	0.2	0.441

Fig. 1. Feature weights in population I are determined by ReliefF.

	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8
I(F;C)	0.24	0.15	0.23	0.3	0.2	0.336	0.2	0.17
Similar(F)	0.15	0.13	0.85	0.66	0.11	0.46	0.1	0.19
W_{IS}	1.6	1.154	0.27	0.455	0.182	0.73	2	0.89
IS_normalize	0.584	0.455	0.2	0.253	0.647	0.332	0.7	0.38

Fig. 2. Feature weights in population II are determined by Mutual Information and feature similarity.

I.

	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8
$W_{R(IS)}$	0.584	0.455	0.2	0.253	0.647	0.332	0.7	0.38

II.

A	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8
X_1	0.23	0.5	0.8	0.24	0.9	0.21	0.42	0.5
X_2	0.9	0.71	0.27	0.12	0.7	0.5	0.3	0.6
X_3	0.24	0.45	0.8	0.51	0.52	0.92	0.42	0.67
X_4	0.79	0.23	0.11	0.86	0.1	0.45	0.45	0.78

III.

P	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8
X_1	1	0	0	1	0	1	1	0
X_2	0	0	0	1	0	0	1	0
X_3	1	1	0	0	1	0	1	0
X_4	0	1	1	0	1	0	1	0

Fig. 3. Initialization strategy and feature mapping relationship of population.

problem, the application of multi-tasking framework in FS is very few. DE-based multi-task algorithms for FS of high-dimensional data need further research.

3. Proposed method

For FS of high-dimensional data, evolutionary algorithms often face the problem of individual stagnation due to the overly broad search scope. As the number of stagnant individuals increases, such algorithms will fall into local optimum and premature convergence.

In this section, a novel multitasking multi-objective differential evolution gene selection algorithm (MMODE) is presented in detail. The principles of MMODE include differentiated populations, elite strategy, multitasking population evolution, and guidance strategy, which are described in section 3.1 to 3.4, respectively. In section 3.5, the overall framework of MMODE is described.

3.1. Differentiated populations

When multitasking differential evolution algorithm is used to solve the optimal solution, the correlation between tasks will have an important influence on the result. MMODE introduces two differentiated populations to enhance the ability to search the optimal subsets in the high-dimensional feature space. Specifically, two populations (corresponding two tasks) are designed to search the same feature space in different directions to increase the diversity of the populations, thus improving the performance of the algorithm.

In the filter phase of MMODE, most irrelevant features are removed by using the ReliefF algorithm, and a promising subset of features is obtained. Next, the weight W_r of features according to ReliefF criterion (*Eq.*(1)) is used for the initialization of population I (task I). And the weight W_{IS} of features according to *Eq.*(13) are used for the initialization of population II (task II).

$$W_{IS}[F_a] = \frac{I(F_a, C)}{Similar(F_a)} \tag{13}$$

where W_{IS} denotes a vector that is used to store the weights of all features, $W_{IS}[F_a]$ denotes weight of a feature F_a, $I(F_a, C)$ is the relevance between feature F_a and the class variable, and $Similar(F_a)$ represents the average similarity between feature F_a and other features, more details can be found in the literature (Moradi & Gholampour, 2016). Specifically, the higher the relevance and the lower the similarity, the more important the feature is.

Before initialization of the task population, the weight W_r of task I and the weight W_{IS} of task II will be normalized to the same interval by *Eq.*(14),

$$W = T_2 * \frac{W - W_{min}}{W_{max} - W_{min}} + T_1 \, W \in \{W_r, W_{IS}\} \tag{14}$$

where W denotes the feature weights of task I and task II, W_{min} represents its lower bound, and W_{max} represents the upper bound. The formula attempts to adjust the feature weights to between $[T_1, T_1 + T_2]$, where $0 < T_1, T_2 < 1$ and $T_1 + T_2 < 1$. The traditional normalization range is between $[0, 1]$, there will be some features whose weight becomes 1 after normalized, causing all individual to select the feature at initialization; or the weight becomes 0, then the feature weight is too low thus causing the feature to be ignored. In this study, T_1 and T_2 were set to 0.2 and 0.5 to scale the value between $[0.2, 0.7]$, which not only ensures that the probability of each feature being selected is positively correlated with its weight, but also avoids the situation that some features are completely selected or excluded. One might ask why the weight values are not mapped to other interval ranges, such as $[0.1, 0.8]$. In fact, this range only ensures that features with lower weights have a lower probability of being selected (but avoid being completely excluded) and features with higher weights have a higher probability of being selected (but avoid necessarily being selected). Our experiments show that there is no significant difference in the experimental results under the conditions of $T_1 \in [0.1, 0.3]$ and $T_1 + T_2 \in [0.5, 0.8]$. The reason should be attributed to the evolutionary mechanism of differential evolution algorithm adopted.

Fig. 1 and Fig. 2 respectively show the calculation principles of the corresponding feature weights of the two populations. The probability of each feature being selected is different in different population individuals because the corresponding feature weight W of each task is different. Moreover, even within the same population, individual chooses features differently due to randomness. If the i th dimension of an individual is initialized to a randomly generated value between 0 and W_i, that is $[0, W_i]$, indicating that the i th feature is selected; otherwise, the i th feature is not selected. The initialization strategies for population I and population II are similar, except that the feature weights are different, resulting in different directions for initializing the populations.

Fig. 3 shows an example of population initialization. Suppose the dataset has 8 features and the number of individuals in the population P is 4. Firstly, weights $W_R(W_{IS})$ is obtained by $Eq.(1)$ (or $Eq.(13)$), and then the weight W is obtained by normalization through $Eq.(14)$. Secondly, a random population $A \in \mathbb{R}^{4*8}$ based on a uniform probability distribution is generated and each element $A_{i,j} \in [0, 1)$. Thirdly, each individual obtains the selected features in the following rule: if $A_{ij} \leq W(F_j)$, then $P_{ij} = 1$ (means feature F_j is selected by individual X_i); otherwise $P_{ij} = 0$ (means feature F_j is not selected by individual X_i). As shown in Fig. 3, the final selected gene subset for individual X_4 is {F2, F3, F5, F7}.

3.2. Elite strategy

In recent years, some scholars have studied multi-objective differential evolutionary (MODE) algorithms based on NSGA-II framework to solve FS problems. However, as the non-domination proceeds, the number of identical individuals increases, which affects the diversity of the population. The selection operator of MODE is greedy selection (as shown in $Eq.(8)$), which selects a better solution between the trial vector and the target vector to survive in the offspring population. If the target vector falls into a local optimum, the generated trial vector is more likely to be worse than the target vector (according to the fitness value). Then the target vector survives into the offspring population, resulting in the parent population and offspring population having the same solution vector. Two identical solution vectors cannot dominate each other, so the two identical solutions survive to the next generation of populations. This reduces population diversity and leads to premature convergence. This means that the non-dominated processes in NSGA-II are not suitable for DE, because DE needs to explore new spaces through vector movements, and too many identical vectors will only limit the performance of DE.

Algorithm 2:Elite Strategy

Input: m: the size of each task population; P: task populations;
Output: E, Elite population;
1: $Es \leftarrow \emptyset$, $E \leftarrow \emptyset$, $i \leftarrow 0$;
2: Calculate the *Score* of individual in task population P by $Eq.(15)$;
3: $P' \leftarrow$ Sorting P in ascending order according to *Score*;
4: Compute the EN by $Eq.(16)$;
5: **while** $len(Es) < EN \& \ i++ < m$ **do**
6: **if** $P'(X_i).HR == Ture$: // X_i is ith individual in population P'
7: $Es = Es \cup P'(X_i)$;
8: **end while**
9: Obtain elite populations E by the **Feature Recombination** (**Algorithm 3**) on Es
10: **return** E;

Algorithm 3:Feature Recombination

Input:Es: subset of excellent individual; ep: elite rate;
L: the probability of weakly correlated features being selected;
Output: E, Elite population;
1:$E \leftarrow \emptyset$;
2: **for** $i = 0; i < len(Es); i++$ **do**
3: Extract selected features of individual $Es(X_i)$ and store them in Fs^i;
4: Compute the vector $MRMR$ of Fs^i by $Eq.(17)$;
5: Convert the $MRMR$ to a binary vector Dif_I according to $Eq.(18)$;
6: **for** $j = 0 to 1/ep$ **do** // $1/ep$ new individuals are generated
7: **new** an individual X;
8: **if** $Dif_I == [1] \& \ [rand()] > [ep]$ **then**
9: The corresponding features are selected by X;
10: **if** $Dif_I == [0] \& \ [rand()] < [L]$ **then**
11: The corresponding features are selected by X;
12: $E = E \cup \{X\}$;
13: **return** E;

To alleviate the above problem, MMODE no longer obtains the next generation population by non-dominated sorting of the offspring population with the parent population. Instead, the offspring population is replaced for non-dominated sorting by elite population. Since the individuals in the elite population are completely different from those in the parent population, there are almost no duplicate individuals in the new population obtained after non-dominated sorting. This effectively enhances the diversity of the population. Therefore, the elite population E is generated by the proposed **Elite Strategy (Algorithm 2)**, which contributes to the solution of the above problem.

In the **Algorithm 2**, elite population E is generated by performing **Feature Recombination** (**Algorithm 3**) based on the elite subset Es. The elite subset Es is the set of the individuals with excellent performance in the parent population P. The performance of individual X is determined by $Eq.(15)$ and whether the individual contains redundant features. $Eq.(15)$ reflects the individual's score on the two minimization objectives of FS. The lower the $Score[X]$, the better the individual.

$$Score[X] = 0.9*F_1(X) + 0.1*F_2(X) \tag{15}$$

In line 6 of the **Algorithm 2**, HR is a Boolean property that indicates whether there is redundancy in the selected features that an individual contains. $HR == Ture$ indicates that the individual contains redundant features, which implies that the individual can be further promoted. The number of EN (obtained by $Eq.(16)$) individuals with $HR == Ture$ are selected from low to high according to $Score[X]$ and stored in the variable Es (line 7).

$$EN = ep*m \tag{16}$$

Where ep represents the proportion of the number of individuals in Es to the population, m represents the number of individuals in each population. An example of the elite subset Es obtained from the parent population P is shown in Fig. 4.

Lines 5–7 of **Algorithm 2** produce an elite subset Es with EN individuals. Then individual X in Es generate $1/ep$ new individuals by the proposed **Feature Recombination (Algorithm 3)**. These new

I.

P	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
f_1	0.12	0.11	0.18	0.09	0.13	0.12	0.16	0.10
f_2	0.45	0.40	0.40	0.41	0.43	0.41	0.46	0.46
Score	0.15	0.14	0.20	0.12	0.16	0.15	0.19	0.14

II.

P	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
Rank	5	3	8	1	6	4	7	2
HR	TURE	TURE	TURE	FALSE	TURE	TURE	FALSE	TURE

III.

P	X_1	X_2	X_3	X_5	X_6	X_8
Rank	5	3	8	6	4	2
HR	TURE	TURE	TURE	TURE	TURE	TURE

IV.

Es	X_2	X_8
f_1	0.11	0.1
f_2	0.4	0.46
HR	TURE	TURE

Fig. 4. An example of elite subset *Es* obtained from the parent population *P*.

individuals are combined into an elite population *E* with the number of individuals *m*.

To avoid the newly generated individuals containing redundant features, in **Feature Recombination (Algorithm 3)**, max-relevance and min-redundancy (mRMR) (Radovic, Ghalwash, Filipovic, & Obradovic, 2017) is used to evaluate the features selected by individuals and classify the features into relevant and irrelevant features. mRMR evaluates features according to the maximum correlation and minimum redundancy criteria. In mRMR, the selected features have maximum correlation with class labels and minimum redundancy with other features. Next, the principle of **Feature Recombination** is described.

In line 3 of the **Algorithm 3**, the selected features in individual $Es(X_i)$ (i th individual of *Es*) are extracted into variable Fs^i. Then, in line 4, the *MRMR* vector of Fs^i is evaluated by *Eq.*(17).

$$MRMR[Fs^i_m] = I(Fs^i_m, Y) - \frac{\sum_{Fs^i_n \in Fs^i} I(Fs^i_m, Fs^i_n)}{|Fs^i|} \tag{17}$$

where $|Fs^i|$ denotes the number of features in Fs^i, $I(Fs^i_m, Fs^i_n)$ represents the mutual information of Fs^i_m and Fs^i_n, Y is the class label, $I(Fs^i_m, Y)$ is the mutual information of feature Fs^i_m and class label Y. Different individuals may contain some of the same features, repeatedly calculating the redundancy between these features is inefficient and a waste of resources. To avoid repeating calculations, in our program, an archive $IS \in \mathbb{R}^{D\times D}$ (D represents the number of features) be designed to store the mutual information of features. If $I(Fs^i_m, Fs^i_n)$ is computed, the value is stored in $IS(Fs^i_m, Fs^i_n)$ to avoid re-computation. For simplicity, the archive variable *IS* does not appear in **Algorithm 3**.

In line 5, the vector *MRMR* is converted into a binary vector according to *Eq.*(18) and stored in the vector *Dif_I*. The value 1 in *Dif_I* indicates that the corresponding feature is important by the individual $Es(X_i)$.

$$Dif_I = \begin{cases} 1 if MRMR > 0 \\ 0 otherwise \end{cases} \tag{18}$$

In lines 7–11, the generation of individuals in the elite population depends on *Dif_I*. The features with *marker* 1 or 0 in *Dif_I* all have the probability of combining to generate a new individual. In lines 8–9, those features with *marker* == 1 and the corresponding value of *rand*() (a random number between 0 and 1) is greater than *ep* are selected. These features well meet the max-relevance and min-redundancy criteria, so they are highly complementary to each other. In lines 10–11, those features with *marker* == 0 and the corresponding value of *rand*() is less than *L* are selected. These features do not meet the max-relevance and min-redundancy criteria, that is, redundant features of the features marked as 0 or the features irrelevant to classification. However, from the perspective of classification accuracy, a small number of redundant features are often helpful to improve classification accuracy and stability, because of the difference between training data and test data. The flowchart of elite strategy is shown in Fig. 5.

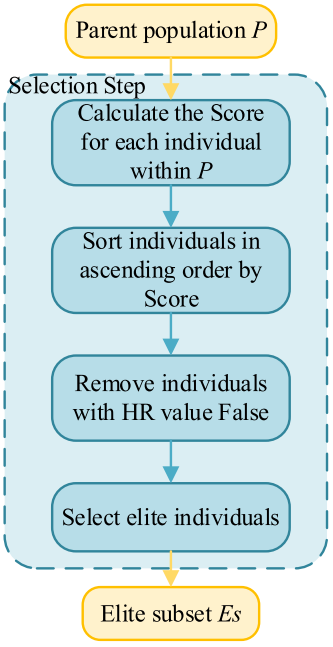


Fig. 5. Flowchart for selecting elite individuals in population *P*.

In a word, each newly generated individual selects more features that conform to mRMR criteria, and a small number of redundant features are selected as supplements. This not only ensures that the generated individuals have better performance, but also ensures the diversity of individuals and increases the effective search.

It should be note that *L* is a small threshold parameter that controls the selection of a small number of redundant features. By setting a larger value of *L*, the global search ability of the algorithm can be enhanced. In the experiments show that $L = 0.01$ is a relatively ideal setting, which

will be discussed and analyzed in detail in the following experimental section.

3.3. Multitasking population evolution

Classical MO-MFEA-II conducts multi-task population evolution according to the task population of the two parent individuals and *RMP* matrix (*Eq.*(12)). MO-MFEA-II is based on genetic algorithm (GA), which generates offspring according to exchange of information between two parent individuals. However, the proposed MMODE is based on the DE algorithm, which generates new individuals based on the spatial motion of a single vector. To satisfy the conditions of multi-task evolution, MMODE is carried out according to the task population of two vectors (i.e., $X_{i,t}$, $X_{best,t}$) and the *RMP* matrix. **Algorithm 4** shows the flow chart of the proposed **Multitasking Population Evolution**.

In line 2 of **Algorithm 4**, the top p% individuals in each task population $P_{k,t}$ ($k \in \{1,2\}$) are saved in archive A_t. In line 5, an individual is randomly selected as $X_{i,t}$ from $P_{k,t}$ and an individual is randomly selected as $X_{best,t}$ from archive A_t. In line 6, the algorithm performs the following operations to generate vector $V_{i,t}$ according to the task population to which the $X_{i,t}$ and $X_{best,t}$ belongs.

If the two individuals $X_{i,t}$ and $X_{best,t}$ belong to the same task population (i.e., $X_{i,t}.k == X_{best,t}.k; k \in \{1,2\}$), then performs **Operation I**.

Operation I: Two individuals $X_{r1,t}$, $X_{r2,t}$ are selected from the task population $P_{k,t}$ where $X_{i,t}$ is located to generate the vector $V_{i,t}$ according to *Eq.*(6). This operation is like single-task population evolution which is mainly used for task population exploitation of the current search direction.

If $X_{i,t}$ and $X_{best,t}$ belong to different task populations (i.e., $X_{i,t}.k \neq X_{best,t}.k; k \in \{1,2\}$), one of the three operations of **Operation II, III and IV** will be executed. Let *rand*_1() denotes a randomly generated number between 0 and 1.

Operation II: If $rand_1() < RMP[X_{i,t}.k, X_{best,t}.k]$, then an individual $X_{r1,t}$ is selected from the task population where $X_{best,t}$ is located and an individual $X_{r2,t}$ is selected from the task population where $X_{i,t}$ is located, to generate the vector $V_{i,t}$ according to *Eq.*(6). This operation generates an individual to explore a new areas because the preconditions for knowledge transfer are met.

If $rand_1() \geq RMP(X_{i,t}.k, X_{best,t}.k)$, one of the two operations of **Operation III** and **IV** will be executed. Let *rand*_2() denotes a randomly generated number between 0 and 1, *ep* is the elite rate.

Operation III: If $rand_2() > ep$, three individuals $X_{r1,t}$, $X_{r2,t}$ and $X_{r3,t}$ are selected from the task population where $X_{i,t}$ is located to generate the vector $V_{i,t}$ according to *Eq.*(5). In the evolutionary search process, if individuals evolve only toward the optimal solution, which tends to fall into a local optimum. Random search is used to reduce this phenomenon according to *Eq.*(5), which improves the ability of the algorithm to explore unsearched regions.

Operation IV: If $rand_2() \leq ep$, two individuals $X_{r1,t}$ and $X_{r2,t}$ are selected from the task population where $X_{best,t}$ is located to generate the vector $V_{i,t}$ according to *Eq.*(6). The evolution of individuals in this operation is also a transfer of knowledge between two tasks.

In line 7, the vector $U_{i,t}$ is generated according to *Eq.*(7). The $U_{i,t}$ vector needs to be decoded into binary values according to *Eq.*(19) before proceeding to the selection operator.

$$U_{bin\,i,j} = \begin{cases} 1 \text{ if } U_{i,j} < W \\ 0 \text{ otherwise} \end{cases}, W \in \{W_r, W_{IS}\} \tag{19}$$

where W is the feature weight of the task corresponding to $X_{i,t}$. The generated new individual $X_{i,t+1}$ performs the same tasks as $X_{i,t}$,

In line 8, according to *Eq.*(20), vector $X_{i,t+1}$ denotes a better vector between vector $U_{i,t}$ ($U_{bin\,i,t}$ is its binary form) and vector $X_{i,t}$ based on their *Score*. In line 9, vector $X_{i,t+1}$ is selected for the next generation P_{t+1}.

$$X_{i,t+1} = \begin{cases} U_{i,t} & \text{if score}(U_{bin\,i,t}) < \text{score}(X_{bin\,i,t}) \\ X_{i,t} & \text{otherwise} \end{cases} \tag{20}$$

Algorithm 4: Multitasking Population Evolution

Input: $P_{k,t}$: t th generation task populations ($k \in \{1,2\}$); p: top p% individuals; m: total number of individuals in both populations; ep: parametric elite rate.
Output: $P_{k,t+1}$, next generation population ($k \in \{1,2\}$);
1: Calculate *RMP* by *Eq.*(12);
2: $A_t \leftarrow$ the top p% individuals of $P_{1,t}$ and the top p% individuals of $P_{2,t}$;
3: $P_{k,t+1} \leftarrow \varnothing$;
4: **for** $i = 0to m$ **do**
5: Select individual $X_{i,t}$ from $P_{k,t}$ and $X_{best,t}$ from A_t; //$k \in \{1,2\}$;
6: Generate the vector $V_{i,t}$ according to the different operations (i.e., **Operations I ~ IV**);
7: Generate the training vector $U_{i,t}$; // crossover operator (*Eq.*(7));
8: $X_{i,t+1} \leftarrow$ the better individual between $U_{i,t}$ and $X_{i,t}$ according to *Eq.*(20);
9: $P_{k,t+1} \leftarrow P_{k,t+1} \cup \{X_{i,t+1}\}$; // $X_{i,t+1}$ performs the same tasks as $X_{i,t}$;
10: **end for**
11: **return** $P_{k,t+1}$;

3.4. Guidance Strategy

For evolutionary algorithms, individuals tend to enter a stagnant state in the process of evolution, which affects the exploration ability of the algorithm. In MMODE, a novel guidance strategy is proposed to alleviate this situation. The proposed guidance strategy uses state detection and feature shrinkage to make individuals jump out of stasis.

To determine whether an individual is in a stagnant state, in MMODE, the individual has an *age* property with an initial value of 0. If an individual stops updating during a certain generation of evolution, its age attribute is increased by 1. If the age of an individual reache threshold *maxAge*, then the individual is in the stagnation stage.

Similar to the **Elite Strategy** (**Algorithm 2**), the guidance strategy also aims to remove redundant features of the individual to achieve better classification performance. For the stagnant individuals, if their attribute $HR == Ture$ (meaning there are redundant features, as in **Algorithm 2**), the **Guidance Strategy** (**Algorithm 5**) removes the features with lower weight from the selected features of the individuals, to make the individuals jump out of the stopped state.

In **Algorithm 5**, X is an input individual with a stagnant state. In line 1, the features selected by the individual X are stored in variable r in ascending order of feature weight. In lines 2–6, the classification error rate of individual X after deleting feature $r(j)$ is evaluated; if the classification error rate is unchanged or is decreased, the algorithm returns the corresponding individual of $X - r(j)$ (i.e., individual X updated successfully and termination guidance strategies); otherwise, the next feature in r is evaluated. Lines 7–8 mean that individual X cannot be updated by guidance strategy, and its property HR is assigned *False* means that the individual no longer participates in **Elite Strategy (Algorithm 2)** and **Guidance Strategy** (**Algorithm 5**).

Algorithm 5:Guidance Strategy

Input: X: individual;
Output: X_{new}, a new individual;
1: $r \leftarrow$ the features selected by the individual X in ascending order of the feature weights W of its task population; // feature weights $W \in \{W_r, W_{IS}\}$
2: **for** $j = 0tolen(r)$ **do**
3: $X_{new} \leftarrow X - r(j)$; // X_{new} represents the corresponding individual after removing //feature $r(j)$ from individual X
4: **if** $f_1(X_{new}) \leq f_1(X)$ **then** // classification error rate
5: **return** X_{new}; // individual X updated successfully
6: **end for**
7: **if** $j == len(r)$ **then** // individual X update failed
8: $X.HR = False$;

In the experimental demonstration (Section 4.2.2), the **Guidance Strategy** can effectively improve algorithm performance. As redundant and irrelevant features in the solution are eliminated, the remaining

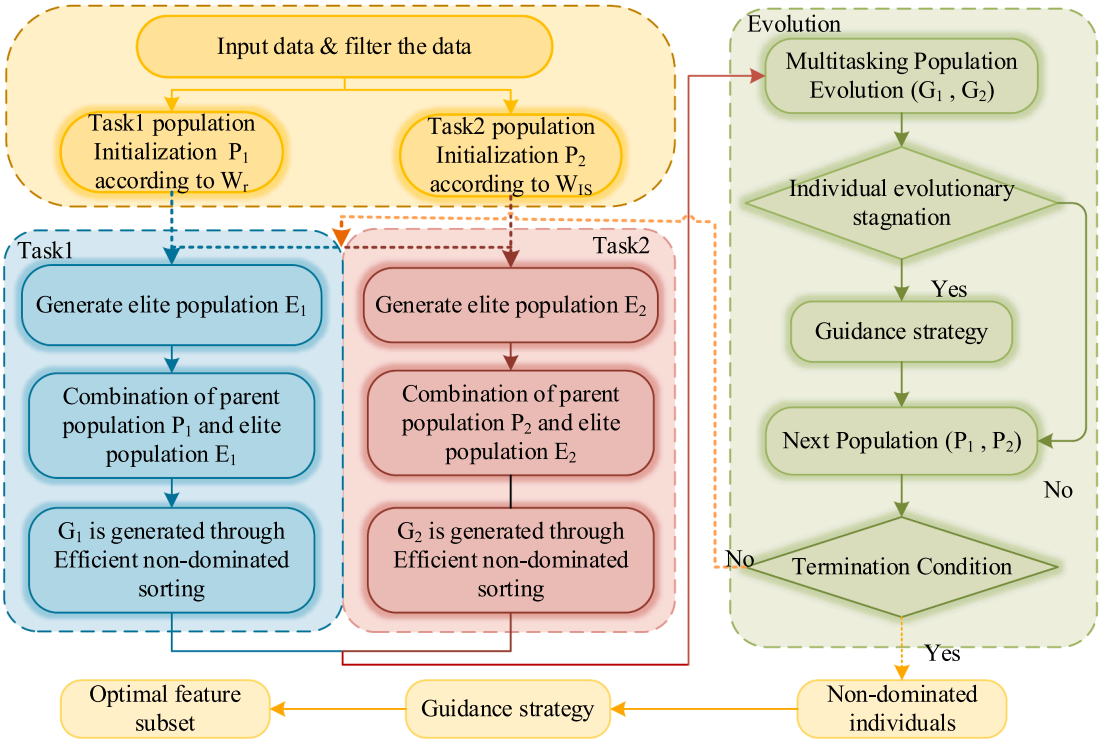


Fig. 6. Flowchart of MMODE.

features tend to perform better.

3.5. Overall framework of MMODE

The proposed MMODE adopts two task populations to search the feature space. Fig. 6 shows the flowchart of the MMODE. As shown in Fig. 5, each task population has its initialization strategy to explore in different regions. After population initialization, the proposed **Elite Strategy (Algorithm 2)** is used to get the elite population E. The population G is generated through the parent population P and the elite population E for efficient non-dominated sorting (Zhang, Tian, Cheng, & Jin, 2015) (P denotes the parent task population, E denotes the elite population generated according to elite strategy, and G is then the result of efficient non-domination of P and E).

After fast non-dominated sorting, the next generation population P is generated through the multitasking evolution of population G (knowledge will transfer between two task populations). The transferred knowledge also affects the distribution of individuals within each task population and the direction of exploration. Once an individual is at a stagnant stage (individuals no longer evolve) in its evolution, then **Guidance Strategy (Algorithm 5)** will help the individual out of this state. When the termination condition is satisfied, the set of non-dominated solutions in both task populations is output. To reduce the influence of redundant features, individuals in the non-dominated solution set will be re-executed to guidance strategy.

The pseudo-code of MMODE is shown in Algorithm 6. In lines 1—4, the two task populations are initialized according to the information entropy weight W_{Is} and reliefF weight W_r of the features, respectively. The evolutionary search process of the two task populations is completed in lines 5–13, which roughly includes three parts. Part I (lines 6–7), two new populations $G_{1,t}$ and $G_{2,t}$ are generated based on the non-dominance sorting of elite populations generated by **Elite Strategy (Algorithm 2)**. Part II (line 8) generates next generation $P_{k,t+1}$ of populations based on multi-task population evolution **(Algorithm 4)**. Part II (lines 9–12), the **Guidance Strategy (Algorithm 5)** is used to update stagnant individuals. After the population evolution is completed, in line 15, the **Guidance Strategy** has performed again on the non-dominant set to remove the redundant features.

Algorithm 6: MMODE

Input: D: training set; m: population size; *maxIterate*: maximum number of generations;
maxAge: the upper limit of the age at which individuals stagnate;
Output: Optimal feature subset;
1: Obtain the weight vector W_r according to *Eq.*(1);
2: Obtain the filtered dataset $D^{'}$ by filtering D according to W_r;
3: Obtain the weight vector W_{Is} based on $D^{'}$ and *Eq.*(13);
4: Initialize the task populations $P_{k,t}$ respectively by using W_r and W_{Is}; //$k \in \{1,2\}$
5: **for** $t = 1$*tomaxIterate* **do**
6: $E_{k,t} \leftarrow$ **Elite Strategy (Algorithm 2)**; //$k \in \{1,2\}$
7 $G_{k,t} \leftarrow$ Efficient non-dominated sorting of $E_{k,t}$ and $P_{k,t}$; //$k \in \{1,2\}$
8: $P_{k,t+1} \leftarrow$ **Multitasking Population Evolution** ($G_{1,t}, G_{2,t}$) **(Algorithm 4)**;
9: **for** each *individual* in $P_{k,t+1}$ //$k \in \{1,2\}$
10: **if** *individual.age* == *maxAge* **then**
11: Update the stagnant individuals on $P_{k,t+1}$ by **Guidance Strategy (Algorithm 5)**;
12: **end for**
13: **end for**
14: Obtain the non-dominated solution set of all task populations;
15: Execute the **Guidance Strategy (Algorithm 5)** on the non-dominated set;
16: **return** Optimal feature subset;

4. Experimental results

To evaluate the overall performance and demonstrate the effectiveness of the proposed algorithm, we compare MMODE with nine popular evolutionary algorithms based on 13 public benchmark microarray datasets. In section 4.1, the experimental framework, including the benchmark datasets, compared algorithms, and parameter settings, are presented. In section 4.2, the experimental results are presented and analyzed.

All experiments are conducted on a PC equipped with Intel Core i5-7300HQ CPU, 2.50 GHz, 16 GB RAM, and Windows 11 64-bit operating system.

Table 1
Description of microarray gene expression datasets.

Dataset	Instances	Feature	Classes
Arcane (Li et al., 2018)	200	10,000	2
BreastCancer1 (Zhu et al., 2007)	97	24,481	2
BreastCancer2 (Chowdary et al., 2006)	104	22,283	2
CLLSUB (Li et al., 2018)	111	11,340	3
CrohnDisease (Burczynski et al., 2006)	127	22,283	3
GLI (Li et al., 2018)	85	22,283	2
Leukemia (Zhu et al., 2007)	72	7129	2
Lung Cancer (Zhu et al., 2007)	181	12,533	2
Ovarian (Zhu et al., 2007)	253	15,154	2
Prostate (Li et al., 2018)	102	12,600	2
SMKCAN (Li et al., 2018)	187	19,993	2
Brain_Tumor (K. Chen et al., 2021)	199	10,367	4
11_Tumor (K. Chen et al., 2021)	174	12,533	11

4.1. Experimental settings

4.1.1. Datasets

All experiments run on eleven publicly available microarray expression profile datasets. The characteristics of these datasets are listed in Table 1, where "Features", "Classes" and "Instances" denote the number of features, the number of classes and the number of samples, respectively.

For a fair comparison, all data are normalized, and all compared algorithms use a uniform filter approach (the top 3 % of genes are selected by using ReliefF) for a fair comparison.

Considering that KNN (Dudani, 1976) is a classically and commonly used classifier and is often used in gene selection tasks (Wang, Xue, Liang, & Zhang, 2022; Wang, Gao, Zhang, & Guo, 2022; Wang et al., 2022), KNN ($K = 10$) is used as the evaluation classifier in the wrapper evaluation stage of all the comparison algorithms for a fair comparison. To avoid large biases, 10-fold cross-validation is adopted to evaluate the accuracy of the gene set obtained by each algorithm. In addition, each comparison method is not a deterministic heuristic algorithm, each of them is run independently 30 times on each dataset, and the average results are then compared to further reduce the impact of the variance.

4.1.2. Compared algorithms

We compare the proposed MMODE algorithm with feature selection algorithms based on evolutionary algorithms. These algorithms include three single-objective algorithms of JADE-NSO (Zeng, Zhang, Chen, & Hong, 2021), HPSO-LS (Moradi & Gholampour, 2016), HLBDA (Too & Mirjalili, 2021), and six multi-objective algorithms of NSGA-II, GR-MOEA (Liu, Chang, & Cheng, 2021), MOBGA-AOS (Xue, Zhu, Liang, & Słowik, 2021), MOFS-BDE (Zhang et al., 2020), CMODE (Pan, Liu, & Chu, 2022), and NMDE (Wang, Xue, Liang, & Zhang, 2022). The brief information on all compared algorithms is as follows.

JADE-NSO: JADE-NSO as an improved algorithm of DE, utilizes the idea of deep search to FS. When an individual is at stagnation, the individual is reset to the historically optimal solution to continue the search.

HPSO-LS: The first stage of HPSO-LS filters out a subset of irrelevant features. Then, the second stage of HPSO-LS utilizes particle swarm optimization (PSO) to search the optimal feature subset from the filtered dataset, in which individuals perform a local search based on feature similarity.

HLBDA: HLBDA is a recently proposed algorithm especially for dealing with gene selection problems based on a binary dragonfly algorithm (BDA). It uses hyper learning strategies to enable BDA to escape local optima and enhance search behavior.

NSGA-II: NSGA-II is one of the most classical multi-objective algorithms and is also used in FS since its efficient performance. Each iteration produces the next population based on the non-dominated sorting of the parent and offspring populations.

GR-MOEA: GR-MOEA is a multi-objective FS algorithm based on the NSGA-II paradigm, which inherits the advantages of NSGA-II. The filter and wrapper populations in the GR-MOEA influence each other during evolution according to guidance and repair strategies.

MOBGA-AOS: MOBGA-AOS is one of the variants of NSGA-II for FS, which uses penalty and reward matrices to evolve individuals from adaptive crossover operators during iteration.

MOFS-BDE: MOFS-BDE is a novel binary multi-objective differential evolutionary algorithm for FS. It improves the one-bit purifying search operator (OPS) and the probability difference-based binary mutation operator, making DE a strong exploration and exploitation capability.

CMODE: CMODE as a pareto-dominance-based multi-objective DE for FS, focuses on ensuring diversity of solutions. CMODE selects elite individuals based on the competition mechanism and elite individuals to guide the evolution of other individuals through a novel mutation operator.

NMDE: NMDE is a multi-objective FS algorithm based on the niche theories. In the framework of evolutionary multi-objective optimization, NMDE uses a mutation operator that combines local and global information of individual nearest neighbor, which effectively speeds up the convergence of the algorithm.

The latter 6 algorithms are all multi-objective algorithms. All of them use efficient non-dominated sorting (Zhang, Tian, Cheng, & Jin, 2015) for multi-objective comparison for the sake of fairness.

4.1.3. Parameter settings

All comparison algorithms use the recommended parameter values suggested in their original papers. The parameter values of MMODE and all compared algorithms are listed in Table 2.

There are two task populations in MMODE and GR-MOEA, the number of individuals in each task population is 70. To fair comparison, in all other eight compared evolutionary algorithms, the population size m is set to 140, which makes all 10 algorithms use the same number of individuals for searching feature subsets. In addition, the maximum number of iterations *maxIterate* is set to 100 for all algorithms. In MMODE, scale factor F is set to 0.5 and the crossover rate (CR) is 0.9.

4.2. Results and analysis

4.2.1. Compared with state-of-the-art algorithms

This section focuses on comparing the MMODE with nine feature selection methods listed in section 4.1.2. Table 3 demonstrates the comparison of all compared algorithms on 13 datasets concerning the Mean Time, Mean Size (the number of selected features), Best Accuracy, and Mean Accuracy (std). The bolded values indicate the best results. The Wilcoxon signed rank test (K. Chen et al., 2021) at the significance level of 0.05 is used to statistically analyze the Mean Accuracy difference between of MMODE and other compared algorithms. In Table 3, the symbols "+", "≈" and "−" respectively indicate that the results of MMODE are better, similar, and worse than the corresponding comparison algorithm.

Table 2
Parameter settings in all comparison algorithms.

Algorithm	Parameter values
JADE-NSO	best individual rate : $p = 0.3$; stagnation tolerance:$T_1 = 8, T_2 = 4$.
HPSO-LS	learning rate:$c_1 = c_2 = 2$; velocity:$v_{max} = 4, v_{min} = -4$
HLBDA	personal learning rate : $pl = 0.4$; global learning rate : $gl = 0.7$.
NSGA-II	$CR = 0.9; MR = 1/featureNum$.
GR-MOEA	$CR = 0.9; MR = 1/featureNum$; strategy execution coefficient : $l = 10$.
MOBGA-AOS	$CR = 0.9; MR = 1/featureNum$; number of repeated : $LP = 5$.
MOFS-BDE	$F = 0.5$*$rand()$; $CR = 0.3$; frequency : $T_{loc} = 5$; turbulence coefficient : $\sigma = 0.01$.
CMODE	$F = 0.5; CR = 0.5$; size of leader set:$\alpha = 5$.
NMDE	$F = 0.5; CR = 0.5$; threshold : $\theta = 0.6$; parameter : $\psi = 1/featureNum$.
MMODE	$F = 0.5, CR = 0.9; p = 0.3; ep = 0.2; L = 0.01; maxage = 3$.

Table 3
Comparison between MMODE and state-of-the-art algorithms on 13 datasets.

Data Sets	Method	Mean Time	Mean Size	Best Accuracy	Mean Accuracy (std)	*W*
Arcane	JADE-NSO	530.69	98.60	96.49	95.90 (0.20)	+
	HPSO-LS	935.47	180.80	93.00	92.00 (0.95)	+
	HLBDA	1078.32	68.40	92.60	91.28 (1.27)	+
	GR-MOEA	715.67	63.40	96.50	94.50 (0.55)	+
	MOFS-BDE	621.34	63.80	97.00	95.60 (0.37)	+
	MOBGA-AOS	604.98	45.80	96.50	95.60 (1.07)	+
	CMODE	551.54	38.00	97.00	96.10 (0.58)	+
	NSGA-II	**463.63**	56.00	96.49	95.00 (0.71)	+
	NMDE	678.25	**16.00**	95.59	94.60 (0.32)	+
	MMODE	898.11	16.40	**98.00**	**97.05 (0.93)**	
BreastCancer1	JADE-NSO	540.69	271.40	95.89	94.18 (1.40)	+
	HPSO-LS	909.95	375.60	91.89	90.69 (0.75)	+
	HLBDA	1007.49	202.30	93.88	92.46 (1.02)	+
	GR-MOEA	606.13	107.60	95.00	94.40 (1.05)	+
	MOFS-BDE	655.31	229.80	93.88	93.19 (0.49)	+
	MOBGA-AOS	466.62	154.40	94.89	93.89 (0.63)	+
	CMODE	**438.80**	124.60	94.88	93.78 (0.94)	+
	NSGA-II	439.85	123.40	94.88	93.89 (1.26)	+
	NMDE	915.15	45.00	93.89	92.51 (0.69)	+
	MMODE	902.35	**32.47**	**97.00**	**96.10 (0.90)**	
BreastCancer2	JADE-NSO	555.32	270.20	100.00	98.09 (0.54)	≈
	HPSO-LS	886.24	582.00	100.00	99.39 (0.43)	≈
	HLBDA	943.56	178.40	100.00	99.03 (0.61)	≈
	GR-MOEA	593.90	65.00	100.00	99.40 (0.49)	≈
	MOFS-BDE	553.07	54.40	99.09	98.67 (0.47)	+
	MOBGA-AOS	458.37	68.00	99.09	98.29 (0.40)	+
	CMODE	**425.50**	46.20	100.00	99.44 (0.46)	≈
	NSGA-II	451.91	22.60	99.09	98.85 (0.38)	+
	NMDE	966.13	16.60	100.00	99.64 (0.45)	≈
	MMODE	815.75	**3.00**	**100.00**	**100.00 (0.00)**	
CLLSUB	JADE-NSO	529.38	113.00	92.80	90.10 (1.80)	≈
	HPSO-LS	766.61	75.67	76.67	75.10 (1.17)	+
	HLBDA	882.32	98.60	77.56	76.18 (1.77)	+
	GR-MOEA	583.17	75.40	91.89	88.58 (3.09)	≈
	MOFS-BDE	657.65	107.60	87.43	83.09 (1.01)	+
	MOBGA-AOS	515.89	71.40	90.07	87.55 (1.03)	+
	CMODE	497.55	80.40	86.51	84.67 (1.53)	+
	NSGA-II	**491.17**	106.60	89.17	87.03 (0.93)	+
	NMDE	669.87	15.40	88.65	87.26 (1.41)	+
	MMODE	762.96	**14.10**	**93.72**	**91.66 (0.75)**	
CrohnDisease	JADE-NSO	562.23	238.20	**99.23**	**98.46 (0.49)**	≈
	HPSO-LS	1065.38	426.33	95.67	95.16 (0.31)	+
	HLBDA	1009.65	187.30	95.11	93.64 (1.72)	+
	GR-MOEA	635.93	82.60	98.43	96.32 (1.22)	+
	MOFS-BDE	592.38	232.20	96.89	95.66 (0.61)	+
	MOBGA-AOS	469.33	184.00	96.32	95.67 (1.17)	≈
	CMODE	451.89	189.80	97.63	96.59 (0.76)	≈
	NSGA-II	**445.06**	168.40	96.86	95.17 (0.75)	+
	NMDE	739.47	78.40	93.88	93.53 (0.03)	+
	MMODE	992.06	**40.80**	99.20	97.48 (0.82)	
GLI	JADE-NSO	522.06	215.40	98.89	98.89 (0.54)	+
	HPSO-LS	886.04	138.67	95.43	94.58 (0.60)	+
	HLBDA	887.54	198.30	95.88	94.26 (0.89)	+
	GR-MOEA	574.21	63.80	98.89	98.09 (0.54)	+
	MOFS-BDE	550.55	175.00	97.65	96.92 (0.59)	+
	MOBGA-AOS	452.96	128.20	97.64	96.42 (0.79)	+
	CMODE	**424.10**	96.80	98.89	97.92 (0.50)	+
	NSGA-II	426.98	75.40	97.78	97.22 (0.63)	+
	NMDE	659.14	28.60	99.01	98.79 (0.05)	+
	MMODE	860.09	**6.10**	**100.00**	**100.00 (0.00)**	
Leukemia	JADE-NSO	484.59	87.30	100.00	100.00 (0.00)	≈
	HPSO-LS	604.34	102.67	100.00	100.00 (0.00)	≈
	HLBDA	659.89	60.40	100.00	99.82 (0.39)	≈
	GR-MOEA	624.73	59.60	100.00	100.00 (0.00)	≈
	MOFS-BDE	513.54	3.40	100.00	100.00 (0.00)	≈
	MOBGA-AOS	436.27	4.80	100.00	100.00 (0.00)	≈
	CMODE	426.85	4.30	100.00	100.00 (0.00)	≈
	NSGA-II	**425.36**	6.40	100.00	100.00 (0.00)	≈
	NMDE	586.54	5.80	100.00	100.00 (0.00)	≈
	MMODE	651.36	**3.00**	**100.00**	**100.00 (0.00)**	
Lung Cancer	JADE-NSO	594.85	120.40	100.00	100.00 (0.00)	≈
	HPSO-LS	874.99	38.33	100.00	99.64 (0.51)	≈
	HLBDA	907.82	69.30	99.01	98.56 (0.99)	+

(*continued on next page*)

Table 3 (*continued*)

Data Sets	Method	Mean Time	Mean Size	Best Accuracy	Mean Accuracy (std)	*W*
	GR-MOEA	853.77	56.00	100.00	99.98 (0.08)	≈
	MOFS-BDE	617.69	35.60	100.00	100.00 (0.00)	≈
	MOBGA-AOS	501.86	20.00	100.00	99.89 (0.21)	≈
	CMODE	472.60	11.80	100.00	100.00 (0.00)	≈
	NSGA-II	**470.96**	8.80	100.00	99.79 (0.26)	≈
	NMDE	789.22	5.00	100.00	100.00 (0.00)	≈
	MMODE	860.80	**3.00**	**100.00**	**100.00 (0.00)**	
Ovarian	JADE-NSO	660.21	156.60	100.00	99.68 (0.16)	≈
	HPSO-LS	1339.24	238.00	100.00	99.73 (0.38)	≈
	HLBDA	1208.65	132.50	100.00	99.13 (0.79)	≈
	GR-MOEA	765.03	56.00	100.00	99.76 (0.19)	≈
	MOFS-BDE	747.87	35.60	100.00	100.00 (0.00)	≈
	MOBGA-AOS	572.91	32.60	100.00	99.87 (0.80)	≈
	CMODE	**517.88**	8.60	100.00	100.00 (0.00)	≈
	NSGA-II	527.22	5.40	100.00	100.00 (0.00)	≈
	NMDE	707.90	3.40	100.00	100.00 (0.00)	≈
	MMODE	958.89	**2.46**	**100.00**	**100.00 (0.00)**	
Prostate	JADE-NSO	475.14	116.40	98.00	98.20 (0.40)	+
	HPSO-LS	845.53	133.33	97.00	96.67 (0.47)	+
	HLBDA	888.65	59.20	97.00	96.43 (0.21)	+
	GR-MOEA	637.50	77.00	99.00	97.96 (0.39)	+
	MOFS-BDE	543.25	54.60	98.00	98.00 (0.10)	+
	MOBGA-AOS	548.19	26.00	98.00	97.80 (0.40)	+
	CMODE	**422.35**	22.80	99.00	98.00 (0.40)	+
	NSGA-II	426.06	17.60	99.00	98.02 (0.04)	+
	NMDE	670.35	17.60	99.00	98.40 (0.49)	≈
	MMODE	730.88	**6.20**	**100.00**	**99.71 (0.37)**	
SMKCAN	JADE-NSO	582.70	208.60	89.77	88.72 (0.03)	+
	HPSO-LS	1084.42	209.33	80.15	79.73 (0.53)	+
	HLBDA	1009.74	124.50	78.99	77.32 (0.76)	+
	GR-MOEA	784.20	78.00	90.79	**89.75 (1.03)**	≈
	MOFS-BDE	743.18	220.00	85.43	83.76 (0.64)	+
	MOBGA-AOS	557.41	190.20	87.13	85.15 (0.97)	+
	CMODE	531.54	143.00	87.13	86.79 (1.99)	≈
	NSGA-II	**510.46**	153.60	87.08	85.33 (0.73)	+
	NMDE	746.73	29.20	86.78	84.65 (1.58)	+
	MMODE	989.19	**27.60**	**90.36**	89.23 (1.18)	
Brain_Tumor	JADE-NSO	335.64	120.43	96.00	92.68 (1.38)	+
	HPSO-LS	416.32	133.22	90.00	91.67 (0.87)	+
	HLBDA	425.32	55.98	90.00	91.48 (0.98)	+
	GR-MOEA	367.45	56.30	96.00	94.19 (1.66)	≈
	MOFS-BDE	342.45	60.32	96.00	93.13 (1.33)	+
	MOBGA-AOS	296.43	50.40	96.00	94.40 (1.40)	≈
	CMODE	**278.34**	46.00	96.00	93.20 (1.27)	+
	NSGA-II	332.76	67.90	92.00	91.27 (1.44)	+
	NMDE	416.23	20.43	94.00	92.93 (1.32)	+
	MMODE	404.23	**18.21**	**98.00**	**97.23 (0.71)**	
11_Tumor	JADE-NSO	304.21	196.47	92.54	91.69 (1.29)	≈
	HPSO-LS	694.22	233.12	88.71	85.94 (1.32)	+
	HLBDA	702.32	112.42	88.45	86.77 (1.52)	+
	GR-MOEA	500.32	104.00	91.23	90.84 (0.63)	≈
	MOFS-BDE	521.34	112.56	92.23	90.78 (0.66)	≈
	MOBGA-AOS	232.42	111.60	92.69	91.44 (0.60)	≈
	CMODE	**209.45**	85.30	**93.09**	**91.82** (1.10)	≈
	NSGA-II	489.23	98.20	92.12	91.01 (0.91)	≈
	NMDE	423.44	20.34	91.68	90.87 (1.03)	≈
	MMODE	598.23	**18.23**	93.00	91.45 (1.20)	

Among all the compared methods, JADE-NSO, HPSO-LS, and HLBDA are single-objective evolutionary algorithms, which regard low classification error rate (high classification accuracy) as the only search objective. Table 3 shows that the three algorithms selected hundreds of gene features for almost every dataset. In contrast, MMODE selects a much smaller number of gene features for each dataset.

Of all the compared methods NSGA-II, GR-MOEA, MOBGA-AOS, MOFS-BDE, CMODE, and NMDE are multi-objective evolutionary algorithms. These algorithms aim to find a set of solutions that tradeoff classification accuracy and the number of features selected. Since many redundant features are excluded, the solution of the multi-objective feature selection algorithm tends to have better performance for classification.

As shown in Table 3, MMODE obtains the highest Best Accuracy on all datasets except for the CrohnDisease and 11_Tumor datasets, obtains the highest Mean Accuracy on all datasets except for the CrohnDisease, SMKCAN, and 11_Tumor, and outperforms all compared methods in terms of the minimum number of selected features on all datasets except the Arcane dataset. In addition, the Mean Accuracy of MMODE reaches 100 % on GLI, Lung cancer, and Ovarian datasets, which has obvious advantages compared with other algorithms, indicating the stability of MMODE. These results prove that MMODE can accurately find key gene subsets for tumor identification.

AUC(Area under the Receiver Operating Characteristics) (Alosaimy et al., 2021) and BACC (Balanced Accuracy) (Li et al., 2020) are two good indicators for evaluating a classifier performance regardless of the severity of class imbalance. To further evaluate the performance of MMODE algorithm on gene selection, the objective function ($f_1(\bullet)$) is

changed as shown in *Eq*.21 and *Eq*.22 respectively.

$$f_1^{'}(X_i) = 1 - AUC(X_i) \tag{21}$$

$$f_1^{''}(X_i) = 1 - BACC(X_i) \tag{22}$$

where $AUC(X_i)$ and $BACC(X_i)$ respectively represent the AUC value and the BACC value obtained by individual X_i. Table 4 demonstrates the comparison of MMODE and other algorithms on 13 datasets concerning the Best AUC, mean AUC (std), Best BACC, and mean BACC (std).

As shown in Table 4, MMODE achieves the highest Mean AUC on all datasets except the 11_Tumor and CrohnDisease datasets, and obtains the highest Best AUC on all datasets except the BreastCancer1 and 11_Tumor datasets. In addition, MMDOE achieves the highest Best BACC and Mean BACC on all datasets except the CrohnDisease and the 11_Tumor datasets. According to the Wilcoxon signed rank test results of Mean AUC (std) and Mean BACC (std), MMODE is superior to other compared algorithms in most cases, which means that MMODE has better performance compared with other algorithms.

Table 5 provides the relative rankings of all the compared algorithms across the eight evaluation criteria. MMODE outperforms the other algorithms in terms of accuracy as well as the number of features relative to each other. MMODE ranks relatively low for Mean Time but achieves a significant advantage in the relative ranking of Mean Times, Mean Size, Best Accuracy, Mean Accuracy, Best AUC, Mean AUC, Best BACC, and Mean BACC. The last row of Table 5 shows the average of the relative rankings for Mean Time, Mean Size, Best Accuracy, Mean Accuracy, Best AUC, Mean AUC, Best BACC, and Mean BACC. MMODE has a great advantage over the other algorithms.

The excellent performance of MMODE is also demonstrated by its ability to achieve better Pareto solutions. Fig. 7 depicts the Pareto front (PF) of 7 multi-objective algorithms on 8 datasets. It is evident from Fig. 7 that the PF of MMODE can dominate the PF of other algorithms. Therefore, MMODE can obtain better non-dominated solutions compared to GR-MOEA, MOFS-BDE, MOBGA-AOS, NSGA-II, CMODE, and NMDE.

4.2.2. Analysis of strategy effectiveness

In this section, we verify the effectiveness of the **Elite Strategy** and **Guidance Strategy** used in MMODE in detail. Specifically, we compare MMODE with MMODE-A, MMODE-B, and MMODE-C, which are all variants of the original MMODE.

MMODE-A: The variant retains the overall framework of MMODE but discards the **Elite Strategy** and the **Guidance Strategy**.

MMODE-B: The variant retains the overall framework of MMODE but does not adopt the **Elite Strategy**.

MMODE-C: The variant retains the overall framework of MMODE but does not adopt the **Guidance Strategy**.

The three MMODE variants above generate offspring populations through multitasking population evolution. Table 6 shows the comparison of experimental results between MMODE and its three variants on 13 datasets concerning the Mean Time (s), Mean Size (the number of selected features), Best Accuracy, and Mean Accuracy.

As can be seen from Table 6, the MMODE-A variant has the worst performance on all datasets in terms of Best Accuracy and Mean Accuracy, except for the Leukemia, Lung cancer, and Ovarian datasets, where the MMODE-A variant performs as well as the other algorithms. In addition, the MMODE-A variant obtains the worst Mean Size (the average number of selected features per dataset is the highest) of the gene subset. Note that the MMODE-A variant does not use the **Elite Strategy** and the **Guidance Strategy**, while the effectiveness of the proposed **Elite Strategy** and the **Guidance Strategy** is verified on all datasets.

Note that the MMODE-C variant uses the **Elite Strategy** but does not use the **Guidance Strategy**, and the MMODE-B variant uses the **Guidance Elite Strategy** but does not use the **Elite Strategy**. The comparison of the experimental results between the MMODE-B variant and the MMODE-C variant allows the analysis of the roles and differences between the **Guidance Strategy** and the **Elite Strategy**. Table 6 shows that compared with MMODE-B, the MMODE-C variant has the better performance on all datasets in terms of Best Accuracy and Mean Accuracy, except for the Leukemia, Lung cancer, and Ovarian datasets, where the MMODE-B variant performs as well as the MMODE-C. In addition, the MMODE-C variant obtains the better Mean Size (the average numbers of selected features for almost datasets are shorter) of the gene subset than MMODE-B. This result indicates that **Elite Strategy** is more effective than the **Guidance Strategy**.

Compared with the **Guidance Strategy** (MMODE-B), although **Elite Strategy** (MMODE-C) achieves better results both in the classification accuracy and the number of selected features, the role of the **Elite Strategy** cannot replace the **Guidance Strategy**. As described in Section 3.4, the **Guidance Strategy** plays a key role in helping the stagnant individuals jump out of the local optima, removing redundant features, and improving the performance of the algorithm. Compared with three variants of MMODE-A, MMODE-B, and MMODE-C, the proposed MMODE combines **Elite Strategy** and **Guidance Strategy**, so it has the advantages of both the **Elite Strategy** and the **Guidance Strategy**. According to Table 6, MMODE obtains the best results both in the classification accuracy and the number of selected features compared with the three variants.

To further show the difference between MMODE and its three variants in the feature selection process, Fig. 8 shows the Pareto front subsets obtained by them on 8 datasets. As shown in Fig. 8, the subset of Pareto fronts obtained by MMODE locates in the bottom left corner of each sub-figure, which means that MMODE selects the least number of features and has the lowest classification error rate. In addition, Fig. 8 intuitively shows that MMODE is better than MMODE-C and MMODE-B, and MMODE-C and MMODE-B are better than MMODE-A. In short, MMODE obtains the smallest gene subset and the best classification performance by using the **Elite Strategy** and the **Guidance Strategy**, which fully proves the effectiveness of the **Elite Strategy** and the **Guidance Strategy**.

4.2.3. Sensitivity analysis of parameters in MMODE

There are three important parameters in the proposed algorithm MMODE, which are *ep* (the elite rate) and *L*(the probability of weakly correlated features being selected) for controlling the **Elite Strategy**, and *maxAge* (the upper limit of the age at which individuals stagnate) for controlling the **Guidance Strategy**. We empirically investigate the effect of parameters *ep*, *L* and *maxAge* on the MMODE.

To obtain the appropriate parameters, we implemented Taguchi's experimental design (Altland, 2006). Taguchi's experimental design is a technique that uses orthogonal arrays to create testing cases that assist in determining the parameters with minimal effort (Gao et al., 2019). The value of the parameter *ep* is chosen from the array $[0.02, 0.08, 0.1, 0.2, 0.3]$, the value of the parameter L is chosen from the array $[0.003, 0.005, 0.01, 0.05, 0.1]$ and the value of the parameter *maxAge* is chosen from the array $[3, 6, 9, 12, 15]$. Therefore, an orthogonal array $L_{25}(5^3)$ was used for experiments to represent the combination of parameter values. The 25 parameter sets are listed in Table 7. To demonstrate the average classification accuracy and the average number of selected features of MMODE with different parameter sets, they are ranked according to Score (*Eq*.15), and the top 5 parameter sets are shown in Table 8. The optimal parameter sets are shown in bold. As can be seen from Table 8, the parameter sets $(ep = 0.2, L = 0.01, maxAge = 3)$ has a Score ranking of *No*.1 for datasets Arcane, BreastCancer1, CLLSUB, and SMKCAN, the *score* ranking on dataset CrohnDisease is *No*.2. According to the results of the parameter discussion, different parameter sets have different effects on the classification performance of MMODE. Therefore, the parameter sets of $ep = 0.2$, $L = 0.01$ and $maxAge = 3$ were adopted by MMODE in our experiments.

Table 4
Comparison of all algorithms under criteria AUC and BACC.

Data Sets	Method	Best AUC	Mean AUC (std)	*W*	Best BACC	Mean BACC (std)	*W*
Arcane	JADE-NSO	97.01	95.71 (0.91)	+	95.71	95.12 (0.31)	+
	HPSO-LS	95.23	94.73 (0.34)	+	92.67	91.87 (0.79)	+
	HLBDA	94.92	94.15 (0.41)	+	92.94	91.91 (0.94)	+
	GR-MOEA	99.11	98.83 (0.19)	≈	94.57	93.79 (0.66)	+
	MOFS-BDE	98.07	97.91 (0.19)	+	95.32	94.91 (0.39)	+
	MOBGA-AOS	99.01	98.74 (0.25)	≈	96.71	95.71 (0.97)	≈
	CMODE	97.55	97.02 (0.21)	+	96.52	95.92 (0.62)	+
	NSGA-II	98.62	98.49 (0.18)	+	95.84	94.65 (1.20)	+
	NMDE	96.26	95.50 (0.63)	+	94.79	94.30 (0.22)	+
	MMODE	**100.00**	**99.23 (0.03)**		**97.88**	**97.39 (0.51)**	
BreastCancer1	JADE-NSO	98.62	98.08 (0.41)	+	95.22	94.41 (0.78)	+
	HPSO-LS	**99.51**	99.16 (0.36)	≈	93.62	92.91 (0.61)	+
	HLBDA	93.50	93.21 (0.31)	+	95.54	94.01 (1.31)	+
	GR-MOEA	95.37	94.40 (1.05)	+	95.61	95.09 (0.49)	+
	MOFS-BDE	97.73	97.19 (0.59)	+	94.83	94.58 (0.28)	+
	MOBGA-AOS	97.28	96.63 (0.53)	+	93.76	93.18 (0.49)	+
	CMODE	95.26	94.56 (0.44)	+	92.65	92.18 (0.48)	+
	NSGA-II	96.61	95.59 (1.14)	+	94.81	94.03 (0.72)	+
	NMDE	95.03	93.51 (0.92)	+	93.65	93.12 (0.34)	+
	MMODE	99.47	**99.29 (0.12)**		**97.66**	**96.85 (0.61)**	
BreastCancer2	JADE-NSO	100.00	99.90 (0.13)	≈	100.00	98.01 (0.41)	≈
	HPSO-LS	100.00	100.00 (0.00)	≈	100.00	99.13 (0.39)	≈
	HLBDA	99.75	99.63 (0.09)	≈	100.00	98.56 (0.54)	≈
	GR-MOEA	100.00	100.00 (0.00)	≈	100.00	98.81 (0.51)	≈
	MOFS-BDE	100.00	100.00 (0.00)	≈	95.85	95.58 (0.27)	+
	MOBGA-AOS	100.00	100.00 (0.00)	≈	98.34	98.12 (0.03)	+
	CMODE	99.93	99.50 (0.24)	≈	98.93	98.72 (0.06)	≈
	NSGA-II	100.00	99.85 (0.20)	≈	98.92	98.13 (0.48)	+
	NMDE	100.00	99.73 (0.21)	≈	100.00	99.23 (0.51)	≈
	MMODE	**100.00**	**100.00 (0.00)**		**100.00**	**100.00 (0.00)**	
CLLSUB	JADE-NSO	95.32	94.22 (1.03)	+	90.27	89.12 (0.79)	≈
	HPSO-LS	85.21	83.10 (1.77)	+	77.13	76.23 (0.81)	+
	HLBDA	84.21	82.88 (1.11)	+	76.10	75.01 (0.90)	+
	GR-MOEA	97.16	96.54 (0.96)	≈	92.12	87.94 (2.27)	≈
	MOFS-BDE	94.71	94.23 (0.23)	+	83.44	82.79 (0.71)	+
	MOBGA-AOS	96.62	96.43 (0.15)	≈	87.62	86.91 (0.63)	+
	CMODE	86.02	85.47 (0.42)	+	85.31	84.86 (0.59)	+
	NSGA-II	94.71	93.41 (1.23)	+	85.84	85.03 (0.77)	+
	NMDE	94.32	94.03 (0.18)	+	88.93	87.96 (0.91)	+
	MMODE	**97.34**	**96.78 (0.26)**		**93.67**	**91.98 (0.98)**	
CrohnDisease	JADE-NSO	97.34	96.57 (0.75)	+	**98.84**	**97.95 (0.61)**	≈
	HPSO-LS	97.21	96.13 (1.01)	+	94.23	92.95 (1.23)	+
	HLBDA	97.19	96.21 (0.92)	+	94.97	93.82 (1.09)	+
	GR-MOEA	100.00	**99.93 (0.06)**	≈	97.33	95.77 (1.52)	+
	MOFS-BDE	98.25	98.18 (0.06)	+	95.89	95.58 (0.27)	+
	MOBGA-AOS	98.35	97.84 (0.37)	+	96.35	94.85 (1.57)	+
	CMODE	97.42	96.59 (0.76)	+	94.69	94.05 (0.63)	+
	NSGA-II	100.00	99.61 (0.17)	≈	94.91	93.97 (1.01)	+
	NMDE	94.38	94.03 (0.18)	+	93.33	92.98 (0.41)	+
	MMODE	**100.00**	99.72 (0.24)		98.31	97.87 (0.48)	
GLI	JADE-NSO	97.01	96.19 (0.74)	+	99.87	98.91 (0.92)	≈
	HPSO-LS	96.78	96.31 (0.52)	+	95.47	94.69 (0.77)	+
	HLBDA	96.41	95.96 (0.42)	+	96.25	94.91 (1.02)	+
	GR-MOEA	99.77	99.63 (0.13)	+	98.58	97.30 (1.48)	+
	MOFS-BDE	99.19	98.37 (0.73)	+	97.69	96.93 (0.82)	+
	MOBGA-AOS	98.11	97.87 (0.19)	+	95.42	93.56 (1.47)	+
	CMODE	97.95	97.76 (0.17)	+	98.29	97.57 (0.80)	+
	NSGA-II	98.31	97.22 (0.63)	+	98.13	97.62 (0.26)	+
	NMDE	100.00	99.63 (0.67)	≈	99.64	98.81 (0.39)	≈
	MMODE	**100.00**	**100.00 (0.00)**		**100.00**	**99.02 (0.04)**	
Leukemia	JADE-NSO	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	HPSO-LS	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	HLBDA	99.92	99.85 (0.03)	+	100.00	99.81 (0.23)	≈
	GR-MOEA	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MOFS-BDE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MOBGA-AOS	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	CMODE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	NSGA-II	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	NMDE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MMODE	**100.00**	**100.00 (0.00)**		**100.00**	**100.00 (0.00)**	
Lung Cancer	JADE-NSO	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	HPSO-LS	100.00	99.94 (0.04)	≈	100.00	98.94 (0.49)	≈
	HLBDA	99.93	99.81 (0.09)	+	100.00	99.05 (0.79)	≈

(*continued on next page*)

Table 4 (*continued*)

Data Sets	Method	Best AUC	Mean AUC (std)	*W*	Best BACC	Mean BACC (std)	*W*
	GR-MOEA	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MOFS-BDE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MOBGA-AOS	100.00	99.91 (0.09)	≈	100.00	99.31 (0.32)	≈
	CMODE	100.00	100.00 (0.00)	≈	100.00	99.24 (0.82)	≈
	NSGA-II	100.00	100.00 (0.00)	≈	100.00	99.14 (0.31)	≈
	NMDE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MMODE	**100.00**	**100.00 (0.00)**		**100.00**	**100.00 (0.00)**	
Ovarian	JADE-NSO	100.00	99.93 (0.16)	≈	100.00	99.71 (0.09)	≈
	HPSO-LS	100.00	100.00 (0.00)	≈	100.00	99.68 (0.25)	≈
	HLBDA	100.00	100.00 (0.00)	≈	100.00	99.16 (0.47)	≈
	GR-MOEA	100.00	100.00 (0.00)	≈	100.00	99.18 (0.17)	≈
	MOFS-BDE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MOBGA-AOS	100.00	100.00 (0.00)	≈	100.00	99.44 (0.35)	≈
	CMODE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	NSGA-II	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	NMDE	100.00	100.00 (0.00)	≈	100.00	100.00 (0.00)	≈
	MMODE	**100.00**	**100.00 (0.00)**		**100.00**	**100.00 (0.00)**	
Prostate	JADE-NSO	97.24	96.45 (0.62)	+	98.00	97.90 (0.16)	+
	HPSO-LS	98.19	97.27 (0.81)	+	96.00	95.92 (0.09)	+
	HLBDA	97.78	97.19 (0.49)	+	97.00	95.97 (0.43)	+
	GR-MOEA	99.47	99.00 (0.51)	≈	98.00	97.65 (0.60)	+
	MOFS-BDE	99.94	99.25 (0.38)	≈	98.00	97.20 (0.40)	+
	MOBGA-AOS	98.93	98.41 (0.40)	+	98.00	96.71 (0.74)	+
	CMODE	97.97	97.73 (0.24)	+	98.00	97.18 (0.04)	+
	NSGA-II	100.00	99.24 (0.48)	≈	99.00	97.62 (0.96)	≈
	NMDE	100.00	99.25 (0.51)	≈	98.00	97.78 (0.51)	+
	MMODE	**100.00**	**99.91 (0.07)**		**100.00**	**99.03 (0.41)**	
SMKCAN	JADE-NSO	94.56	93.64 (1.05)	+	89.23	87.62 (0.33)	+
	HPSO-LS	87.29	86.18 (0.81)	+	79.01	77.89 (0.66)	+
	HLBDA	85.52	84.61 (0.91)	+	79.38	78.02 (0.61)	+
	GR-MOEA	96.53	95.71 (1.01)	≈	90.22	88.65 (1.41)	≈
	MOFS-BDE	93.88	93.53 (0.53)	+	82.40	81.71 (0.41)	+
	MOBGA-AOS	94.44	94.29 (0.31)	+	83.39	82.61 (0.53)	+
	CMODE	88.17	87.86 (0.39)	+	87.23	86.97 (0.41)	+
	NSGA-II	92.59	92.21 (0.53)	+	86.26	83.28 (0.61)	+
	NMDE	92.41	91.66 (1.05)	+	87.13	84.68 (1.24)	+
	MMODE	**96.57**	**96.06 (0.45)**		**90.41**	**88.91 (0.71)**	
Brain_Tumor	JADE-NSO	99.51	99.23 (0.12)	≈	93.00	92.38 (0.92)	+
	HPSO-LS	99.33	99.06 (0.20)	≈	91.00	90.99 (0.65)	+
	HLBDA	99.49	99.18 (0.18)	≈	92.00	90.78 (1.21)	+
	GR-MOEA	100.00	99.72 (0.26)	≈	96.00	94.06 (1.29)	≈
	MOFS-BDE	99.79	99.53 (0.11)	≈	93.00	92.83 (0.41)	+
	MOBGA-AOS	99.83	99.41 (0.21)	≈	94.00	92.00 (1.79)	+
	CMODE	100.00	100.00 (0.00)	≈	93.00	92.17 (0.82)	+
	NSGA-II	99.53	99.11 (0.24)	≈	92.00	91.02 (0.53)	+
	NMDE	100.00	100.00 (0.00)	≈	95.00	92.03 (1.22)	+
	MMODE	**100.00**	**100.00 (0.00)**		**97.00**	**95.94 (1.01)**	
11_Tumor	JADE-NSO	100.00	99.69 (0.24)	≈	91.03	90.09 (0.72)	≈
	HPSO-LS	88.52	87.35 (0.12)	+	86.64	85.92 (0.83)	+
	HLBDA	88.21	87.46 (1.05)	+	87.96	86.89 (1.21)	+
	GR-MOEA	99.88	99.71 (0.09)	≈	91.32	90.24 (1.18)	≈
	MOFS-BDE	99.53	99.49 (0.05)	≈	90.96	90.58 (0.28)	≈
	MOBGA-AOS	99.61	99.43 (0.15)	≈	**92.49**	**91.58 (0.97)**	≈
	CMODE	**100.00**	**100.00 (0.00)**	≈	91.26	90.16 (0.93)	≈
	NSGA-II	99.40	99.09 (0.35)	≈	91.33	90.71 (0.69)	≈
	NMDE	99.29	99.08 (0.13)	≈	91.49	90.91 (0.81)	≈
	MMODE	99.36	99.23 (0.15)		92.42	91.55 (0.91)	

Table 5
Relative rank of overall methods on 8 criteria.

Criteria	JADE-NSO	HPSO-LS	HLBDA	GR-MOEA	MOFS-BDE	MOBGA-AOS	CMODE	NSGA-II	NMDE	MMODE
Mean Time	3.77	9.00	9.70	6.62	5.23	3.07	**1.54**	1.84	6.92	7.31
Mean Size	8.46	9.38	7.31	5.23	6.77	5.23	4.39	4.54	2.62	**1.08**
Best Accuracy	4.42	8.15	8.31	4.34	6.04	5.39	4.54	5.69	5.58	**2.54**
Mean Accuracy	3.32	8.77	9.38	5.00	5.73	5.19	4.50	5.58	5.70	**1.92**
Best AUC	5.50	7.00	9.42	3.73	4.85	4.62	6.12	4.92	5.77	**3.08**
Mean AUC	6.15	7.31	9.08	3.50	4.19	4.85	5.77	5.69	5.92	**2.54**
Best BACC	4.35	8.15	7.04	4.12	6.54	5.46	6.00	5.69	5.27	**2.38**
Mean BACC	4.00	8.46	8.69	4.08	5.31	6.23	5.54	5.85	5.08	**1.77**
Average	5.00	8.28	8.62	4.58	5.58	5.01	4.80	4.98	5.36	**2.83**

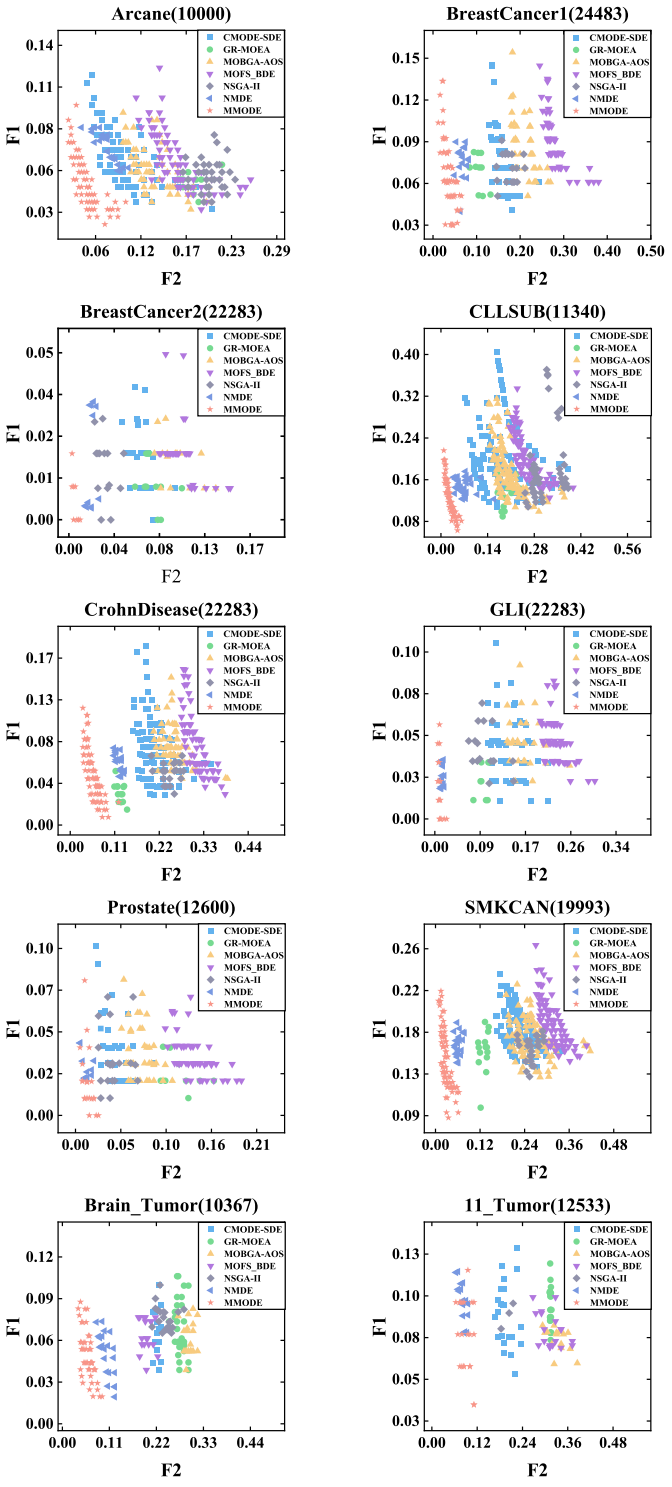


Fig. 7. The Pareto fronts obtained by applying CMODE, NSGA-II, MOBGA-AOS, MOFS-BDE, NMDE, GR-MOEA, and MMODE. F1 denotes the classification error rate, F2 denotes the proportion of the number of selected features.

4.2.4. Time analysis

This section compares the running time of MMODE with nine feature selections based on evolutionary algorithms listed in section 4.1.2 and three variants of MMODE listed in 4.2.2.

As can be seen from Table 3, MMODE has shorter running time than single-objective algorithms (JADE-NSO, HPSO-LS and HLBDA) in most cases. In the wrapper evaluation stage, the feature subset should be evaluated by cross validation mode. The computation time of KNN is

Table 6
Comparison between MMODE and its three variants.

Data Sets	Method	Mean Time	Mean Size	Best Accuracy	Mean Accuracy (Std)	*W*
Arcane	MMODE-A	544.24	38.60	94.00	93.30 (0.60)	+
	MMODE-B	611.10	31.40	96.00	94.90 (0.37)	+
	MMODE-C	764.20	23.40	97.50	96.80 (0.81)	+
	MMODE	898.11	**16.40**	**98.00**	**97.05 (0.93)**	
BreastCancer1	MMODE-A	553.70	146.40	92.74	91.59 (0.40)	+
	MMODE-B	622.55	53.40	94.89	94.04 (1.49)	+
	MMODE-C	758.15	54.20	97.89	95.28 (1.02)	≈
	MMODE	902.35	**32.47**	**97.00**	**96.10 (0.94)**	
BreastCancer2	MMODE-A	463.90	37.80	100.00	99.43 (0.47)	≈
	MMODE-B	531.31	11.00	100.00	100.00 (0.00)	≈
	MMODE-C	642.24	11.20	100.00	100.00 (0.00)	≈
	MMODE	815.75	**3.00**	100.00	**100.00 (0.00)**	
CLLSUB	MMODE-A	393.69	40.20	80.16	80.81 (1.44)	+
	MMODE-B	462.06	33.80	87.39	85.26 (0.88)	+
	MMODE-C	532.50	18.20	92.80	90.49 (1.06)	≈
	MMODE	762.96	**14.10**	**93.72**	**91.66 (0.75)**	
CrohnDisease	MMODE-A	497.17	128.20	91.38	89.66 (0.93)	+
	MMODE-B	655.28	103.00	94.49	93.14 (0.90)	+
	MMODE-C	811.42	74.00	98.43	96.59 (0.79)	≈
	MMODE	992.06	**40.80**	**99.20**	**97.48 (0.82)**	
GLI	MMODE-A	419.67	69.80	98.82	97.41 (0.88)	+
	MMODE-B	608.25	28.00	100.00	98.35 (0.58)	+
	MMODE-C	645.43	20.20	100.00	100.00 (0.00)	≈
	MMODE	860.09	**6.10**	100.00	**100.00 (0.00)**	
Leukemia	MMODE-A	365.25	29.20	100.00	100.00 (0.00)	≈
	MMODE-B	427.36	25.60	100.00	100.00 (0.00)	≈
	MMODE-C	507.65	4.00	100.00	100.00 (0.00)	≈
	MMODE	651.36	**3.00**	100.00	**100.00 (0.00)**	
Lung Cancer	MMODE-A	434.17	18.60	100.00	100.00 (0.00)	≈
	MMODE-B	631.33	11.20	100.00	100.00 (0.00)	≈
	MMODE-C	750.69	3.40	100.00	100.00 (0.00)	≈
	MMODE	860.80	**3.00**	100.00	**100.00 (0.00)**	
Ovarian	MMODE-A	510.78	9.20	100.00	100.00 (0.00)	≈
	MMODE-B	671.79	5.20	100.00	100.00 (0.00)	≈
	MMODE-C	816.52	3.00	100.00	100.00 (0.00)	≈
	MMODE	958.89	**2.46**	100.00	**100.00 (0.00)**	

(*continued on next page*)

Table 6 (*continued*)

Data Sets	Method	Mean Time	Mean Size	Best Accuracy	Mean Accuracy (Std)	*W*
Prostate	MMODE-A	354.85	34.00	98.00	97.63 (0.46)	+
	MMODE-B	448.04	13.80	99.00	98.00 (0.40)	+
	MMODE-C	580.32	12.60	100.00	99.61 (0.48)	≈
	MMODE	730.88	**6.20**	100.00	**99.71 (0.37)**	
SMKCAN	MMODE-A	412.87	67.80	80.74	79.55 (0.38)	+
	MMODE-B	634.84	53.60	84.43	84.28 (0.97)	+
	MMODE-C	827.54	30.60	88.80	87.88 (1.11)	+
	MMODE	989.19	**27.60**	**90.12**	**89.23 (1.18)**	

larger if the selected feature subset includes more features (KNN needs to calculate the distance between samples). Obviously, MMODE is more likely to select a smaller feature subset compared to the single objective algorithm.

As shown in Tables 3 and Table 5, MMODE takes slightly longer to run compared with the multi-objective algorithms. MMODE focuses on selecting a few of genes to achieve high classification accuracies, it adopts a new elite strategy to calculate feature relevance and uses a new guidance strategy that requires repeated evaluation of the classification performance of feature subsets. Therefore, MMODE requires more computational resources. However, the running time of all comparison algorithms is in the same order of magnitude. On the whole, the calculation time of MMODE does not increase significantly compared with that of competitors, but it achieves the best classification performance and the smallest gene subset.

5. Conclusion

In the classification of microarray gene expression profile data, existing multi-objective evolutionary algorithms are prone to local optimality in feature (gene) selection. In this study, a new gene selection algorithm MMODE is proposed, aiming to select a few genes in tumor classification and achieve higher accuracy. The main contributions of the proposed MMODE include the following aspects. (1) An efficient initialization strategy is proposed to effectively differentiate between task populations. The strategy enables specifying the search direction of the different task populations. (2) The final feature subset with higher quality is obtained based on the knowledge transfer of the two task populations. (3) MMODE uses a new elite strategy and a new guidance strategy to reduce the search space and avoid falling into local optima.

The effectiveness of MMODE is validated by comparing it with state-of-the-art algorithms on a 13-gene microarray dataset. The experimental results demonstrate that MMODE is a promising approach for solving gene selection problems. Because MMODE shows better or at least competitive performance in terms of classification accuracy and number of selected genes compared with other algorithms. Moreover, based on their reported results, MMODE also shows superiority over other evolutionary algorithms from the literature. All these suggest that the proposed elite and guiding strategies can optimize the search space of the population, improve the quality of individuals in the population, and help the algorithm to escape from local optima.

Some further studies related to MMODE will be carried out. First, we will extend MMODE to solve other real-world problems in domains with multiple optimal solution requirements. Second, we continue to optimize and improve the time consumption and data space occupation of MMODE. Finally, this study confirms that the method of co-evolution of two differentiated task populations can help to find a better solution to

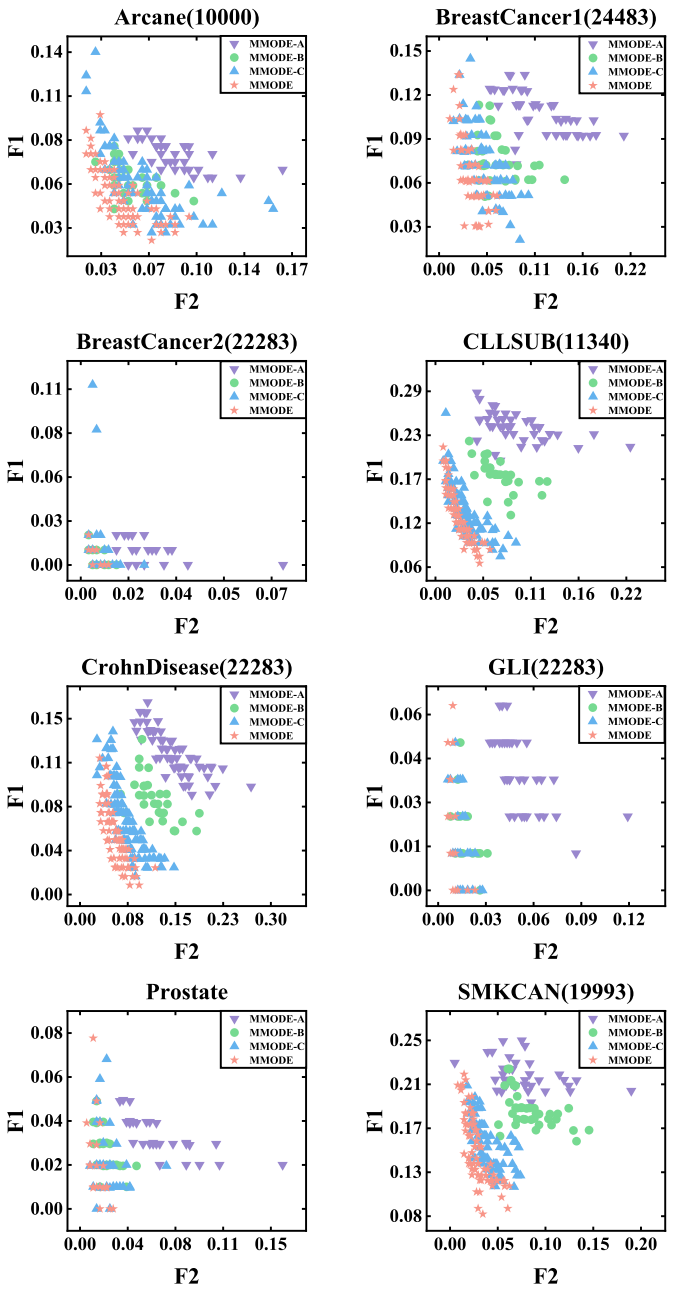


Fig. 8. The Pareto front subsets obtained by MMODE-A, MMODE-C, MMODE-B, and MMODE. F1 denotes the classification error rate, F2 denotes the proportion of the number of selected features.

Table 7
Parameter settings of experiments.

No.	*ep*	*L*	*maxAge*	No.	*ep*	*L*	*maxAge*
1	0.02	0.003	3	14	0.1	0.05	3
2	0.02	0.005	6	15	0.1	0.1	6
3	0.02	0.01	9	16	0.2	0.003	12
4	0.02	0.05	12	17	0.2	0.005	15
5	0.02	0.1	15	18	0.2	0.01	3
6	0.08	0.003	6	19	0.2	0.05	6
7	0.08	0.005	9	20	0.2	0.1	9
8	0.08	0.01	12	21	0.3	0.003	15
9	0.08	0.05	15	22	0.3	0.005	3
10	0.08	0.1	3	23	0.3	0.01	6
11	0.1	0.003	9	24	0.3	0.05	9
12	0.1	0.005	12	25	0.3	0.1	12
13	0.1	0.01	15				

Table 8
Result of parameter comparison in MMODE.

Data Sets	No.	*ep*	*L*	*maxAge*	Mean size	Mean Accuracy (Std)	Score
Arcane	1	**0.2**	**0.01**	**3**	**17.40**	**97.03** (0.83)	**0.0325**
	2	0.08	0.01	3	19.34	97.02 (0.84)	0.0331
	3	0.2	0.003	12	21.27	97.07 (0.76)	0.0335
	4	0.02	0.01	9	23.23	96.98 (0.84)	0.0349
	5	0.1	0.05	3	22.72	96.49 (0.77)	0.0391
BreastCancer1	1	**0.2**	**0.01**	**3**	31.64	**96.14** (0.84)	**0.0453**
	2	0.1	0.1	6	32.55	96.10 (0.94)	0.0460
	3	0.1	0.05	3	34.72	96.13 (0.92)	0.0464
	4	0.08	0.003	6	34.55	96.10 (0.67)	0.0466
	5	0.2	0.05	6	**31.55**	95.95 (0.89)	0.0470
CLLSUB	1	**0.2**	**0.01**	**3**	14.10	**91.65** (0.75)	**0.0797**
	2	0.1	0.05	3	14.23	91.45 (0.46)	0.0817
	3	0.08	0.003	6	15.10	91.36 (0.73)	0.0827
	4	0.02	0.005	6	15.13	91.36 (0.77)	0.0828
	5	0.1	0.05	3	15.26	91.26 (0.69)	0.0837
CrohnDisease	1	**0.02**	**0.003**	**3**	**40.58**	**97.56** (0.86)	**0.0355**
	2	0.2	0.01	3	40.88	97.45 (0.66)	0.0365
	3	0.02	0.05	6	41.38	97.46 (0.66)	0.0366
	4	0.08	0.005	9	41.72	97.48 (0.66)	0.0366
	5	0.2	0.01	3	41.83	97.43 (0.66)	0.0370
SMKCAN	1	**0.2**	**0.01**	**3**	27.60	**89.23** (1.16)	**0.1061**
	2	0.1	0.05	6	27.93	89.19 (1.09)	0.1066
	3	0.1	0.003	9	27.25	89.14 (1.38)	0.1068
	4	0.2	0.05	6	28.57	89.15 (1.21)	0.1071
	5	0.1	0.005	12	**26.73**	89.02 (1.18)	0.1077

the feature selection problem, but how to control more populations to search for the optimal solution is also an important research task. One of our next research tasks is to investigate whether using three or more task populations is more beneficial for problem solving.

CRediT authorship contribution statement

Min Li: Conceptualization, Investigation, Writing – review & editing. **Yangfan Zhao:** Conceptualization, Methodology, Visualization. **Mingzhu Lou:** Data curation. **Shaobo Deng:** Data curation, Software. **Lei Wang:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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